



Co-DIRECT: A knowledge-augmented multi-agent framework for interactive drama script generation

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ABSTRACT

Drama script generation is a challenging yet valuable task with applications in film, gaming, and digital storytelling. Current LLM- and agent-based methods achieve some narrative coherence but often suffer from content homogenization, limited creativity, and underdefined human roles.

To address these issues, we propose **Co-DIRECT**, a human-AI co-creation framework that positions a human director as a central collaborator in a *Director-in-the-Loop* (DiL) paradigm and employs a multi-agent structure: a Writer Agent constructs narrative arcs, an Actor Agent performs context-aware dialogues, and a Critic Agent evaluates consistency and emotional expressiveness. An ontology-driven knowledge graph extracts narratological principles from classical scripts to guide plot development and iterative refinement.

Extensive evaluations show that Co-DIRECT outperforms strong baselines in narrative coherence, emotional depth, content diversity, and script structure. These results highlight the benefits of combining narratological knowledge, multi-agent collaboration, and DiL human guidance for controllable and creative script generation.

1. Introduction

Similar to screenwriting (Chen et al., 2024) or storytelling (Kumaran et al., 2024), drama script generation (DSG) is a specialized form of story creation (Han et al., 2024), involving long-form text generation with strong demands on causality and logical progression (Yao et al., 2019). Advances in large language models (LLMs) have opened new possibilities for DSG, enabling the automated production of structurally complex and engaging scripts.

A defining feature of DSG is that the linguistic behavior and actions of characters must align with their predefined personalities, and character development must be well-structured. One milestone in agent-based behavioral simulation is Generative Agents (Park et al., 2023), which simulates human-like behavior in a sandbox world by assigning LLM agents various social roles. Despite these advances in character-level simulation, the lack of explicit narrative direction often results in stories that drift away from intended dramatic arcs. This highlights the need for frameworks that can balance local improvisation with global narrative control.

Beyond character-centric simulations, prior research on DSG can be broadly categorized into two types of approaches: *progressive* and *hierarchical*. Progressive methods expand narratives iteratively based on a given theme and prior context. For example, *AI Dungeon* (Latitude, 2019) prompts user intervention after each step to revise, continue, or end the story. While this allows dynamic development, it often leads to coherence degradation in longer scripts and lacks autonomous control over planning and termination. To address these issues, hierarchical methods adopt a top-down structure. Yao et al. (2019) first generate a high-level storyline to guide coherent expansion. Likewise, *Co-Writing* (Mirowski et al., 2023) organizes generation into multiple layers—titles, character bios, plot outlines, scene summaries, and dialogues. Despite improved vertical consistency, these systems still face coherence loss in lower levels, especially between plot events and character dialogue as scripts grow longer. However, the rigid top-down design can limit flexibility and reduce emergent diversity in character behavior, highlighting a trade-off between global structural control and adaptive narrative development.

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Building upon these progressive and hierarchical methods, recent work has further explored multi-agent frameworks that coordinate specialized roles to enhance narrative coherence, improve controllability, and reduce the burden of manual prompting. For instance, IBSEN (Han et al., 2024) introduces a Director Agent to outline the major plot, guide agent behavior, and ensure temporal flow. Similarly, HoLLMwood (Chen et al., 2024) employs a Writer–Editor pairing, where the Editor provides feedback on character arcs and plot structures. Agents’ Room (Huot et al., 2025) orchestrates a larger group of specialized agents via a central controller, reducing the manual effort of crafting complex prompts.

While multi-agent methods provide a richer simulation of collaborative authorship, they still leave the human role underdefined. Existing approaches to human–AI co-creation in drama script generation face two critical limitations. First, in current human–LLM co-creation systems, the division of labor between human creators and AI models is often unclear. As demonstrated by works such as LAMP (Chakrabarty et al., 2025) and Dramatron (Mirowski et al., 2023), humans are frequently relegated to the role of content reviewers or intermediaries in information transfer. This places a heavy burden on humans to monitor, filter, and stitch together model outputs, diverting effort away from the genuinely creative aspects of scriptwriting. Second, even when multi-agent architectures introduce role specialization inspired by real-world workflows, the position of the human participant remains narrowly defined. For instance, DRAGN (Ashby et al., 2023), IBSEN (Han et al., 2024), and interactive script generation frameworks (Wu et al., 2025) cast humans primarily as “players,” enhancing immersion but depriving them of the director’s fine-grained narrative control. Similarly, Agents’ Room (Huot et al., 2025) situates the human solely as a system-level coordinator of agent orchestration, with little involvement in the artistic dimension of script creation. In all these cases, the human is present, but not in the director’s seat. This misalignment with real-world drama practices, where the director coordinates multiple contributors, ensures consistency, and provides artistic direction, limits both creative agency and narrative control. To alleviate these limitations, we propose the concept of *Director-in-the-Loop* (DIL), which explicitly establishes the human director as a central collaborator who provides culturally grounded background materials, configures narrative intent, supervises iterative refinement, and intervenes in agent interactions. This formulation restores the human director’s role as both a creative authority and a structural component of the collaborative generation process.

Despite progress across progressive, hierarchical, and multi-agent frameworks, current systems remain constrained by templated prompts, leading to content homogenization, a common challenge reported by users in practical DSG scenarios (Tian et al., 2024). Recent empirical studies (Tang et al., 2025) revealed that both professional screenwriters and hobbyists encounter low narrative diversity due to repeated structural patterns. Furthermore, while LLMs can exhibit imaginative output in science fiction or surreal genres, both the diversity and logical plausibility of generated content decline significantly in realistic storylines, prompting some users to avoid AI-generated scripts for such themes. This phenomenon is partially attributed to what has been termed the algorithmic monoculture (Chakrabarty et al., 2025), an inherent tendency of LLMs to replicate dominant patterns from their training data, which may limit creativity in open-ended generation tasks like DSG. Although LLMs possess the capacity to hallucinate novel content, they often lack grounding in specific real-world or cultural contexts. Balancing *coherence*, *diversity*, and *control* remains a critical research challenge.

A promising direction is knowledge-enhanced generation, which augments pretrained LLMs with structured external knowledge to support coherent and contextually grounded plot generation. For example, Wang et al. (2023a) demonstrates the effectiveness of structured knowledge integration in generating long-form and coherent narratives, especially in domain-specific storytelling scenarios. In a related effort, Pan et al. (2025) embeds previously generated plot events into a dynamic knowledge graph, which functions as a form of evolving memory to guide subsequent story development. However, most of these

approaches focus primarily on commonsense or domain-level knowledge and seldom incorporate narratological or dramaturgical knowledge derived from classic scripts. Knowledge-enhanced generation offers a promising avenue, yet existing methods typically rely on manually authored knowledge graphs (Ashby et al., 2023) that emphasize plot content or commonsense, rather than narratological structure or culturally grounded context. Furthermore, how such knowledge enhancement strategies contribute to improving narrative creativity and diversity remains insufficiently explored. This work therefore addresses the following core research question: how can we design a human–AI collaborative framework that integrates narratological knowledge to improve coherence, controllability, and creative diversity in drama script generation?

To address the resulting challenges in narrative coherence, controllability, and creative diversity, we propose **Co-DIRECT** (Co-creation with Director-WRiter-Actor-critic), a multi-agent framework that embeds dramatic principles within role-based collaboration to improve coherence, controllability, and human–AI co-creativity in drama script generation. As illustrated in Fig. 1, the Writer Agent constructs narrative arcs and designs character biographies; the Actor Agent performs context-aware improvisational dialogues based on scene-level cues; the Critic Agent evaluates the generated content against the intended plotline, assessing narrative consistency and emotional expressiveness. These agents operate in a stage-wise, director-guided workflow that closely emulates the collaborative dynamics of professional screenwriting practice.

The key contributions of this work are as follows:

- We propose an ontology-based knowledge modeling approach tailored for drama script generation, which systematically extracts narratological and dramaturgical elements from classic scripts and organizes them into a structured knowledge graph.
- We design **Co-DIRECT**, a collaborative multi-agent framework that emulates real-world screenwriting workflows through the coordinated roles of a Writer, Actor and Critic, embedded within a Director-in-the-Loop paradigm that places the human director at the center of the creative process.
- We establish a multi-dimensional evaluation protocol and conduct extensive experiments against strong baseline systems. Results demonstrate that Co-DIRECT outperforms prior methods in narrative coherence, emotional authenticity, and content diversity.

2. Related work

2.1. Multi-agent frameworks in DSG

Although large language models (LLMs) have shown remarkable performance in general text generation, they struggle with the intricacies of long-form drama scriptwriting. In particular, maintaining narrative coherence and consistent character portrayal across multiple scenes remains challenging. These difficulties largely stem from the limited modeling capacity of single-agent systems and the lack of fine-grained coordination mechanisms for managing dynamic character behaviors and evolving plot structures.

To overcome these challenges, recent work has turned to multi-agent paradigms. One line of research simulates open-ended virtual environments, where autonomous agents with persistent identities engage in self-directed expression and social interaction. For instance, sandbox-based systems (Park et al., 2023) foster emergent behavior through long-term unscripted enactments, while *The Drama Machine* (Magee et al., 2024) enhances character depth by incorporating psychoanalytic constructs such as ego and superego. While effective at modeling spontaneity and internal complexity, such approaches often lack structural guidance and controllability, rendering them less suited for drama script generation, which demands thematic focus, narrative progression, and authorial alignment.

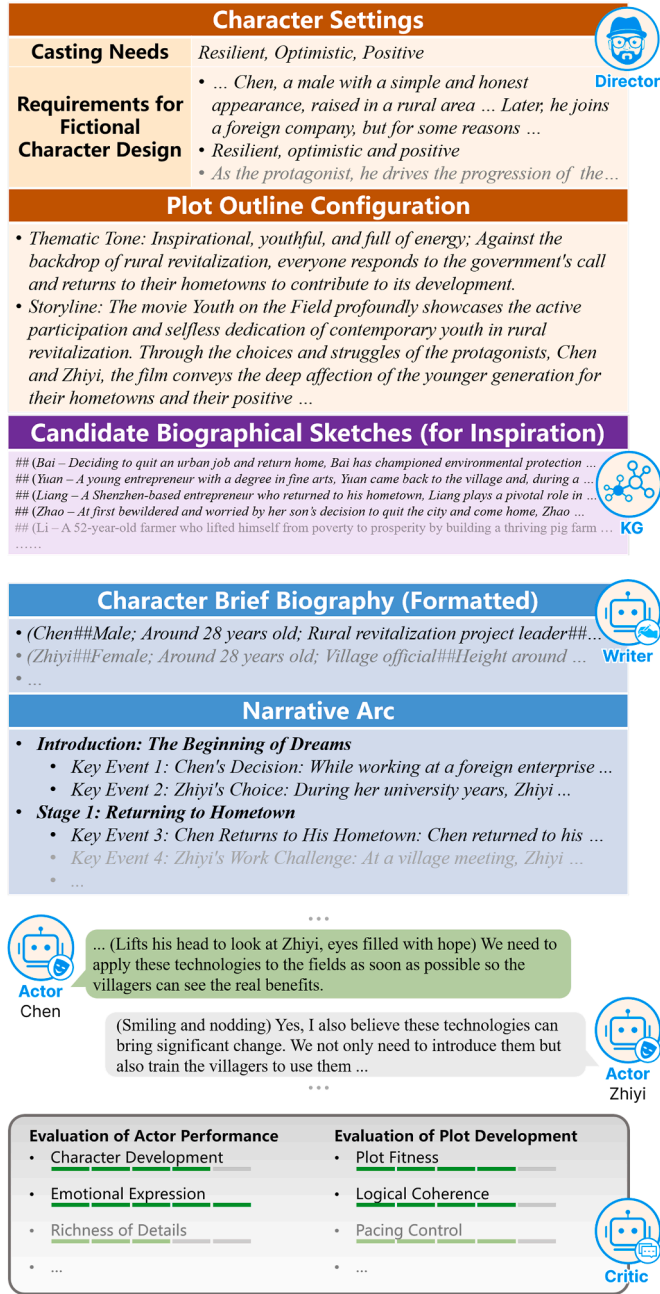


Fig. 1. Co-DIRECT is a knowledge-enhanced multi-agent co-creation framework that utilizes agents' collaboration to generate drama scripts that conform the creative intent of human directors.

Building on the principle of task decomposition, recent work has shown that dividing the DSG task into subtasks handled by specialized agents can lead to improved generation quality. A more structured alternative emulates real-world screenwriting workflows by assigning role-specific responsibilities to multiple agents. For example, wr-AI-ter (Weber et al., 2024) decomposes the generation process into stages such as ideation, structure, refinement, and export. Similarly, MetaGPT (Hong et al., 2023) organizes agent collaboration through predefined team roles; HoLLMwood (Chen et al., 2024) and AGENTS' ROOM (Huot et al., 2025) simulate collaborative writers' rooms; and IBSEN (Han et al., 2024) emphasizes director-actor coordination to maintain consistency between character goals and dramatic actions.

Despite offering improved modularity and structural transparency, existing frameworks often adopt rigid pipelines and rely on fixed agent

behaviors, limiting their adaptability and creative flexibility. Human input is typically confined to the initial configuration stage, with few opportunities for real-time steering or iterative refinement.

To address these limitations, our proposed Co-DIRECT framework introduces a human-in-the-loop architecture supported by an ontology-driven knowledge base and three specialized agents: Writer, Actor, and Critic. This design enables interpretable and co-creative workflows that combine human guidance with agent specialization, bridging the gap between structural control and expressive diversity in DSG.

2.2. Recent SOTA in drama script generation

Recent advances in drama script generation (DSG) can be broadly situated within two methodological paradigms: *progressive* and *hierarchical*. Progressive approaches expand narratives iteratively, where each step conditions on the previously generated content and user or agent interventions. Examples include interactive systems such as *AI Dungeon* (Latitud, 2019), which extends stories turn by turn, and recent work on chain-of-thought (CoT) narrative planning (Senanayake & Ware, 2025), where an LLM is used as a heuristic guide to predict next actions or estimate remaining steps toward a narrative goal. These predictions inform classical narrative planners such as Sabre (Ware et al., 2021), enabling step-by-step action generation conditioned on the current world state and character beliefs. While this approach improves local decision-making, it does not construct high-level plot structures such as outlines or story graphs, and can suffer from inconsistency, world-state drift, and domain-specific failure. Multi-agent simulations such as *Generative Agents* (Park et al., 2023) and *The Drama Machine* (Magee et al., 2024) also fall under this paradigm when they emphasize emergent behaviors developed through stepwise interactions rather than global structural guidance. While progressive methods are flexible and support improvisation, they often struggle to preserve coherence in long scripts due to the lack of overarching control.

Hierarchical methods, in contrast, adopt a top-down decomposition strategy that structures generation across multiple levels. Representative systems include *Dramatron* (Mirowski et al., 2023), which organize script creation into layered subtasks such as outlines, summaries, and dialogues. Recent frameworks like *wr-AI-ter* (Weber et al., 2024) and multi-agent pipelines such as *HoLLMwood* (Chen et al., 2024), *IBSEN* (Han et al., 2024), and *Agents' Room* (Huot et al., 2025) also exemplify this paradigm by assigning role-specific responsibilities (e.g., writer, editor, director) and coordinating outputs across stages. These systems improve vertical consistency and enable more structured authoring, but may still face mismatches between high-level planning and scene-level dialogue.

It is important to note that most of these progressive and hierarchical methods operate in a fully automated manner, providing flexibility but leaving the human role underdefined. As a result, they may struggle to maintain narrative coherence, character consistency, or creative diversity over long scripts. Our proposed Co-DIRECT framework addresses this limitation through a DiL paradigm, which will be discussed in detail in Section 2.4.

Overall, progressive and hierarchical methods collectively capture the dominant state-of-the-art (SOTA) in DSG. By abstracting our experimental baselines into these two paradigms, we ensure that the evaluation is representative of both incremental expansion and structured decomposition strategies, while remaining aligned with recent SOTA developments.

2.3. Knowledge-enhanced script generation

Knowledge-enhanced generation aims to improve coherence, factuality, and depth by incorporating structured or retrieved knowledge into the generation process. In narrative generation, such techniques have employed resources such as commonsense graphs (e.g., ConceptNet Liu & Singh, 2004), story graphs Valls-Vargas et al., 2017, and manually

annotated knowledge bases (e.g., DRAGN Ashby et al., 2023) to provide contextual grounding and guide text generation.

In the context of drama, prior work has explored the use of character descriptions, plot templates, or narrative scripts to guide generation. For example, structured *prompts chain* (Mirowski et al., 2023) containing character and plot elements was utilized to help LLMs generate consistent and plausible scenes. Jackson and Latham (2022) can be adapted to different stories and characters, enabling the AI-controlled roles to fit into a variety of narrative scenarios. These approaches have shown promise in improving short-term consistency but often fall short in maintaining long-range narrative coherence and complex character development across multi-scene scripts.

Despite recent progress, most methods rely on shallow or linear knowledge structures, failing to capture the hierarchical dependencies among dramatic roles, emotional arcs, and character development. They also neglect performative elements, such as speech style and interaction dynamics, crucial for engaging scripts. While some incorporate retrieval, few encode narratological knowledge in reusable, interpretable forms.

We draw on classical narratological theories. Freytag's Pyramid (Bucher et al., 1975) provides the classic dramatic structure of setup, confrontation, and resolution, while the Hero's Journey (Campbell, 2008) outlines archetypal stages of a protagonist's transformation, from the call to adventure to the return with newfound wisdom. Polti (1917) defines possible types of conflicts and plot-driving events, Archer (1913) summarizes narrative techniques such as suspense, twist, irony, and symbolism, and Field (2005) offers guidance on arranging scenes within the three-act structure. These theories were selected as they represent well-established frameworks in narratology and screenwriting, widely used in drama studies and creative writing pedagogy. We encode these narrative structures and techniques into our script corpus, forming a structured and interpretable representation of dramatic knowledge. These encoded elements not only provide guidance for script generation but also enhance interpretability, ensuring that generated scripts exhibit coherent plots, well-developed characters, and engaging narrative arcs.

Building on this narratologically informed corpus, we further organize the encoded elements into a drama-specific knowledge graph via ontology-driven extraction from exemplar scripts. This structured resource bridges the gap between shallow templates and interpretable narratological knowledge and supports both planning and critique: the Writer Agent uses it to build coherent, character-aligned plots, and the Critic Agent evaluates thematic and emotional consistency. This enables structurally grounded yet creatively expressive generation.

2.4. Narrative ontology modeling

Knowledge-enhanced generation has shown promise in improving narrative quality through structured knowledge integration. However, most existing methods for drama script generation rely on raw-script learning via fine-tuning or prompting (Li et al., 2023; Shao et al., 2023; Wang et al., 2023b), which limits their ability to handle complex narrative tasks involving long-term consistency and emotional depth.

Ontology-based modeling addresses this challenge by formalizing key narrative elements, including characters, events, goals, and emotions, into structured and interpretable representations that support reasoning and planning. Previous works such as PersonaBank (Lukin et al., 2016) exemplify *Story Intention Graphs* approach in the context of narrative understanding, although they have rarely been applied to script generation. Jupybara (Wang et al., 2025) is a data analysis and storytelling tool that introduces a design space framework emphasizing the integration of domain knowledge with real-world contexts to produce actionable insights. This approach has inspired us to extract narratological knowledge from classic scripts and apply it to DSG to enrich the semantic precision, cultural context, and thematic depth of generated scripts.

Several systems implicitly reflect ontological thinking. From Role-Play to Drama-Interaction uses narrative chains for scene progression.

Yan and Tang (2023) incorporates temporal information into event chains, while Waseem et al. (2023) additionally considers the creator's emotions within the event structure. Yet, these lack formal integration of narratological semantics into generative systems.

We propose an explicit ontology-driven framework that extracts structured knowledge as triples and organizes them into a drama-specific knowledge graph. This supports generation and evaluation in our Co-DIRECT system: the Writer agent uses it for narrative planning, the Actor for dialogue generation, and the Critic for coherence assessment. This integration improves interpretability, control, and alignment with human creative workflows.

2.5. Human-AI co-creation in scriptwriting

The integration of human creativity and AI capabilities has become a growing focus in scriptwriting research. While LLMs can generate fluent and stylistically consistent content, they often lack the nuanced understanding of dramatic structure, character psychology, and thematic coherence that experienced human creators provide. As such, recent studies have explored human-AI co-creation paradigms that embed human control into various stages of the scriptwriting process.

Early systems like Dramatron (Mirowski et al., 2023) adopt a co-writing workflow, where human writers provide initial inputs and review AI-generated outputs. Evaluations show that such systems can assist with idea generation and structure, but still face issues such as logical inconsistencies, shallow character arcs, and stereotyped content. Similarly, *1001 Nights* (Sun et al., 2022) demonstrates a storytelling interaction loop in which a human player and an AI agent alternately develop narratives, highlighting both the creative potential and coordination challenges of such collaborations.

More recent agent-based frameworks attempt to structure this collaboration. IBSEN (Han et al., 2024) emphasizes director-actor interactions for goal-driven dialogue generation. Wu et al. (2025) propose an interactive drama framework focusing on immersive experience and player agency. However, these systems typically limit human involvement to initial planning or post-hoc editing, lacking mechanisms for real-time intervention, fine-grained control, or iterative refinement during generation.

Beyond scriptwriting, human-AI co-creation has also been systematically examined in visual design and multimodal creativity. For instance, CrePoster (Zhang et al., 2024) integrates expert knowledge into aesthetic rule definition and iterative evaluation, forming a feedback loop between AI generation and human judgement. Cui et al. (2025) embeds expert insights into model design, dataset construction, and evaluative feedback, while Cui et al. (2022) demonstrates iterative jury-based labelling to refine retrieval quality. Similarly, MLF (Liu et al., 2023) combines expert-assisted annotation with human scoring of AI-generated captions, highlighting how human input grounds cultural and metaphorical interpretation. Although these studies arise from the art and design domains, their methodological insights, particularly the systematic integration of human experts at multiple stages, are transferable to narrative co-creation.

In contrast, our proposed Co-DIRECT framework emphasizes a more specific paradigm: a *Director-in-the-Loop* (DiL) design. Compared to the broader notion of human-in-the-loop, this framing highlights the specialized, central role of the director as both orchestrator and evaluator of the creative process. Within multi-agent systems, the DiL paradigm positions the human director as a central collaborator who configures agent roles, supervises their interactions, and ensures alignment with narrative intent. By guiding refinement and intervening when necessary, the director functions not only as a creative authority but also as an organizing force that maintains coherence and cultural depth across agent contributions. The human director initiates the process through multi-dimensional configurations and retains the ability to intervene at any stage. Co-DIRECT thus enables controllable and co-creative generation workflows that mirror real-world scriptwriting practices. This is

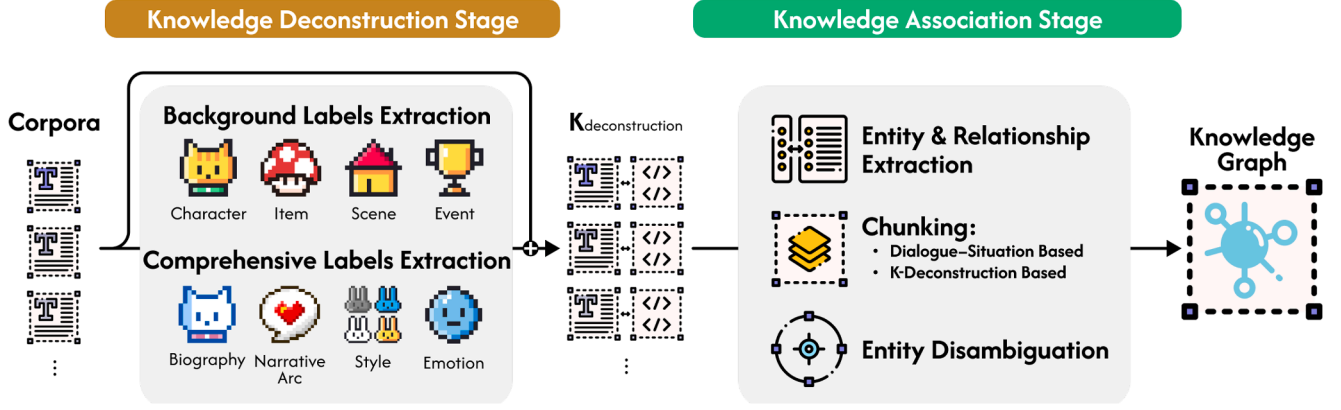


Fig. 2. Two-stage knowledge-base construction diagram.

achieved by coordinating three specialized agents, namely the Writer, the Actor, and the Critic, under the director's guidance.

3. Narrative-aware ontology for creative support

One of our objectives is to provide human directors with valuable creative references derived from canonical scripts. However, existing LLM-based script generation tools tend to produce homogeneous outputs with limited diversity and weak inspirational power, which falls short of this objective. At the root of this limitation is the fact that current models largely rely on the recombination of their internal, finite repertoire of dramatic knowledge, without sustained narrative coherence across longer contexts. Moreover, these homogeneous outputs also reflect inherent biases in pretrained LLMs, which may reinforce stereotypes or favor certain cultural and stylistic patterns.

Although advanced tuning and retrieval techniques (e.g., SFT or RAG) can enhance model performance on specific corpora, they typically capture only surface-level textual patterns. In our preliminary practice, we observed that essential narrative elements, such as character dossiers, narrative arcs, scene-specific details, and critical plot cues, remain underexplored by such methods. This issue becomes even more salient when multiple reference scripts of similar themes or cultural backgrounds are available: cross-script entity association emerges as a crucial factor for generating diverse and stimulating creative insights.

To address these challenges, we propose a narratology-informed paradigm for knowledge graph construction that integrates narrative theory with ontology-based representation. This paradigm aims to transform unstructured scripts into structured, interconnected knowledge that can directly support human creative tasks. Specifically, as shown in Fig. 2, our method consists of two stages: (i) a knowledge deconstruction stage, which extracts and structures implicit dramatic elements, and (ii) a knowledge association stage, in which the extracted narrative elements are linked as entities within a knowledge graph and aligned across scripts. The following subsections elaborate on these two stages in detail.

3.1. Knowledge deconstruction stage

Inspired by classical narratological theories, we design a chained, hierarchical knowledge extraction method that leverages the inductive capabilities of LLMs to extract a series of increasingly deep implicit semantic layers from script texts. These include background knowledge tags, character biographies, narrative arcs, and artistic style labels, which together provide a semantic foundation for subsequent ontology modeling and structured representation. The semantic deconstruction process proceeds as follows.

1. **Background Knowledge Tagging.** Identifies key elements such as characters, props, settings, events, and narrative clues in the script, organizing them into “tag–description” pairs.
2. **Character Biography Construction.** Constructs character profiles by aggregating descriptions across scenes, including personality traits, behavioral tendencies, social relationships, and dramatic functions. These biographies reflect both explicit information and inferred attributes from the dialogue and plot context.
3. **Narrative Arc Construction.** By integrating the character biographies and background tags, extracts key stages of plot development and constructs a complete narrative structure (e.g., beginning, conflict, climax, reversal, flashback, and resolution).
4. **Artistic Style Analysis.** Provides a descriptive analysis of the script's overall linguistic style, narrative mode, and emotional tone, serving as supplementary semantic information.

The outputs from each of these steps are standardized into structured tag sets with strong computational interpretability. Through this method, rich implicit semantic information can be extracted from raw scripts, serving as enhanced contextual input for downstream tasks such as entity recognition, relationship extraction, and entity disambiguation in knowledge graph construction.

3.2. Knowledge association stage

3.2.1. Problem definition and notation

The target in knowledge association stage is aim to transform the complex semantics within scripts and related corpora into structured entity–relationship–entity triplets (e_i, r_j, e_k) , thereby enhancing the representational and reasoning capabilities of the knowledge graph.

Let the corpus be $D = \{d_1, d_2, \dots, d_p\}$. The set of entities extracted from D is $E = \{e_1, e_2, \dots, e_n\}$, and the set of relations is $R = \{r_1, r_2, \dots, r_m\}$. The triplet set is defined as:

$$T = \{(e_i, r_j, e_k) \mid e_i, e_k \in E, r_j \in R\}$$

Each triplet represents a semantic fact, reflecting the connection between different elements in the script. The relevant notations are defined as follows.

- **Document segment** d_p denoted as $d_p = \{text_p, index_p\}$, where $text_p$ is the textual content of the script segment and $index_p$ is its index.
- **Entity** e_i corresponds to a key element in the script (e.g., character, setting, event, item), represented as $e_i = \{id_i, type_i, desc_i\}$, indicating the entity identifier, entity type, and text description, respectively.
- **Relationship** r_j represented as $r_j = \{id_j, type_j, desc_j\}$, similarly containing a unique identifier, relationship type, and textual description.

Based on these definitions, structured knowledge triplets can be extracted from each document segment, denoted as:

$$T = \bigcup_{d_p \in D} \{(e_i, r_j, e_k) \mid e_i, e_k \in E_p, r_j \in R_p\}$$

where E_p and R_p are the sets of entities and relations extracted from segment d_p , respectively. The resulting triplet set serves as the backbone for constructing a knowledge graph that is both structurally complete and semantically coherent.

In practice, we abstracted two optimization strategies, each formalized with explicit notations: a document chunking strategy and an entity disambiguation strategy.

3.2.2. Document chunking strategy

A proper document chunking strategy is critical for improving the quality of triplet extraction. Since script texts are typically lengthy, they must be divided into semantically coherent and structurally independent segments to facilitate entity and relationship recognition by the model.

Existing works on knowledge extraction typically rely on fixed-size sliding windows with overlap to segment long texts. While simple to implement, this strategy often disrupts the natural structure of the document: semantic units may be split apart, and truncated tokens at the boundaries can be incorrectly identified as entities. Such fragmentation introduces noise and undermines the accuracy of the resulting knowledge graph.

To this end, we design two complementary strategies, denoted collectively as:

$$S = \{S_{\text{dialogue}}, S_{\text{deconstruct}}\}$$

Applying S to the raw corpus D_{raw} yields a segmented document set:

$$D = \text{Split}(D_{\text{raw}}, S)$$

The two strategies are described as follows:

- **Dialogue–Situation Based Chunking.** Instead of cutting by fixed window size, we preserve complete situational units (e.g., a continuous block of dialogue, a scene description, or a narrative paragraph) as the basic segmentation granularity. This ensures that contextually bound entities remain intact and semantically coherent.
- **Knowledge-Deconstruction Based Chunking.** For explicit narrative knowledge obtained in the deconstruction stage (e.g., background notes, character biographies, narrative arcs), we treat each label and its associated description as a segmentation boundary. This preserves the integrity of explicit knowledge units and avoids cross-boundary fragmentation.

Together, these strategies better align segmentation with the intrinsic narrative structure of scripts and the semantic units of deconstructed knowledge, thereby providing more reliable input for subsequent entity extraction and relationship association.

3.2.3. Entity disambiguation strategy

The process of extracting entities and relations from a document segment d_i is defined as:

$$E = \text{ExtraEntity}(d_i),$$

$$R = \text{ExtraRelationship}(d_i, E_a, E_b)$$

In practice, script texts often exhibit two types of ambiguity: (1) the same entity may appear under multiple aliases across different contexts, and (2) the same mention may refer to different entities in different contexts. These issues motivate the need for entity disambiguation to ensure that the constructed knowledge graph remains accurate and consistent.

To this end, we introduce $K_{\text{deconstruct}}$ as an *explicit knowledge base*, where each element (l_j, k_j) represents a label–description pair. This resource serves as the pivotal bridge between context-dependent mentions and canonical entities in the graph. By aligning an entity's local context

to labels in $K_{\text{deconstruct}}$, we anchor mentions to a standardized representation, thereby mitigating aliasing and referential inconsistencies.

Formally, let the context of an entity e_i be c_i with vector representation $\vec{c}_i$. The matching score with label l_j is given by:

$$\text{Match}(e_i, l_j) = \frac{\text{Sim}(c_i, k_j)}{\sum_{k \in K} \text{Sim}(c_i, k)},$$

where $\text{Sim}(X, Y)$ denotes semantic similarity in the embedding space. A higher score indicates stronger alignment with the canonical label.

While this local score captures pairwise affinity, it does not reflect the entity's position relative to the entire knowledge set. We therefore compute a normalized vector of matching scores:

$$\vec{m}_i = [\text{Match}(e_i, l_1), \text{Match}(e_i, l_2), \dots, \text{Match}(e_i, l_{|K|})].$$

This global representation offers twofold benefits: on the one hand, the normalization imposes a global constraint that alleviates the impact of individual annotation errors; on the other hand, placing all entities in a shared vector space enables clustering and visualization. The highest-scoring label provides the canonical anchor for disambiguation, while the vector form further allows researchers to inspect ambiguous regions and intervene when necessary.

Finally, if the similarity between two matching vectors satisfies $\text{Sim}_{\text{vec}}(\vec{m}_i, \vec{m}_k) \geq T_{\text{merge}}$, the corresponding entities e_i and e_k are merged and an equivalence relationship r_{sameas} is recorded.

4. Co-DIRECT framework with DiL

Co-DIRECT addresses four long-standing limitations in automatic script generation: incoherent plots, weak human controllability, severe content homogenization, and the lack of domain-grounded knowledge integration, through an integrated design. As illustrated in Fig. 3, the system adopts a multi-agent co-creation architecture that mirrors the division of labor in professional screenwriting teams, distributing narrative development, performative realization, and critical evaluation across Writer, Actor, and Critic Agents to enhance coherence and diversity. To ensure stronger human involvement, Co-DIRECT further incorporates a *Director-in-the-Loop* (DiL) mechanism that elevates the director from a passive reviewer to an active authority capable of steering narrative intent, providing cultural grounding, and intervening in agent interactions. Complementing these components, an ontology-driven knowledge base serves as a shared memory that encodes structured cultural, historical, and narrative elements, thereby maintaining coherence, supporting traceability, and enabling the director to inject domain expertise throughout the workflow.

The following subsections elaborate on the workflow structure, the roles of the three agents, and their interaction with both the director and the knowledge base.

4.1. Co-creation workflow

The Co-Creation Workflow in Co-DIRECT formalizes collaboration among multiple specialized agents under the supervision of a **Director-in-the-Loop (DiL)**. Unlike conventional human-in-the-loop settings, DiL represents an expert-level intervention, providing targeted guidance across narrative planning, character design, dialogue generation, and creative evaluation. This ensures that agent outputs meet structural and stylistic goals while retaining creative flexibility.

A shared **knowledge graph (KG)** functions as the central repository of narratological knowledge, character archetypes, and event templates. It supports agents in retrieving relevant context and maintaining consistency, while also allowing the DiL to oversee decisions and refine outcomes.

This section introduces a five-stage workflow: knowledge base construction, character design, narrative arc planning, improvisational dialogue generation, and critique with revision. At each stage we highlight the responsible agents, the use of the knowledge graph, and the points

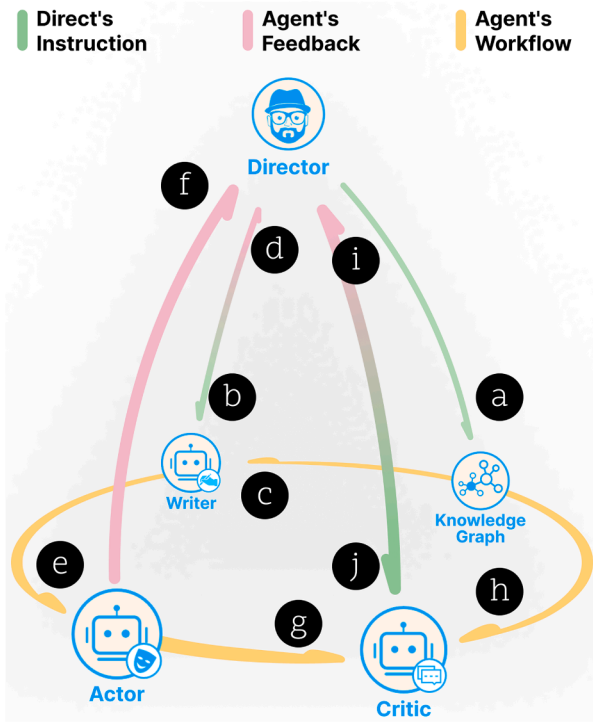


Fig. 3. A workflow diagram illustrating the collaboration among roles within the Co-DIRECT framework: the Director provides corpora and then builds up the Knowledge Graph (KG) (a); the Writer Agent (WA) receives character settings and plot outlines from the Director (b) and, in parallel, retrieves creative labels from the KG (c); then it returns the designed fictional characters and narrative arc (d); The Actor Agent (AA) receives this narrative design (e), then generates improvised multi-turn dialogues and immediately delivers them to the Director (f); the complete script, including narrative arc and dialogues, is submitted to the Critic Agent (CA) (g); Optionally, the creative labels was retrieved as suggestions for script refinement (h); the CA then delivers performance evaluations to the Director (i) and the Director oversees the CA's iterative assessment and revision process (j).

where the DiL can intervene to guide or adjust the process. Through this design, Co-DIRECT establishes a structured, iterative collaboration that mirrors professional screenwriting practices while integrating automated agents with expert oversight.

4.1.1. Stage I: knowledge base construction

Prior to generation, the Director provides exemplar drama scripts or reference documents that capture narratological knowledge, artistic styles, and specific cultural contexts. These raw materials are first deconstructed to extract explicit narratological elements (e.g., plot structures, character archetypes, key events) and cultural background information, which are formalized as structured labels. The extracted labels are then integrated into a knowledge graph grounded in narratological principles. The KG serves as the shared memory for all agents and the Director, ensuring that subsequent character design, narrative planning, and dialogue generation remain traceable, explainable, and guided by enriched narrative knowledge. This deconstruction step enables the system to go beyond surface-level textual features, providing a principled foundation for improved narrative coherence, creativity, and diversity.

4.1.2. Stage II: new character design

This stage involves two sub-processes: character selection and synthesis. First, the system retrieves related character archetypes from the knowledge graph, guided by the initial narrative specifications provided by the Director. These archetypes, annotated with personality traits, social functions, and dramatic roles, serve as reference candidates. The

Director may emphasize particular cultural archetypes or narrative functions to be prioritized in this retrieval process. Then, the system composes new fictional characters by combining features from a subset of the candidates, ensuring diversity in character profiles while allowing the DiL to refine or approve the final roster.

4.1.3. Stage III: narrative arc planning

Based on the designed characters and retrieved motif patterns, the Writer Agent constructs narrative structures, typically employing either the five-act structure or the “Hero’s Journey” model. The five-act structure divides the plot into five phases: Exposition, Rising Action, Climax, Falling Action, and Resolution. The “Hero’s Journey” model provides a character-driven trajectory that outlines a hero’s transformative adventure. The system also anticipates more complex designs in the future, such as non-linear storytelling, reflecting the agent’s flexibility and expressive potential. At this stage, the Director may review and revise the generated outline, guiding the narrative arc to align with thematic intentions, pacing requirements, or cultural constraints informed by the knowledge graph.

4.1.4. Stage IV: improvisational dialogue generation

In this stage, the Actor Agents decompose each narrative arc node into performable shots and generate multi-turn dialogues for each scene. Dialogue is driven primarily by the agents’ interactions, allowing spontaneous, character-consistent exchanges. The Director intervenes only when necessary to halt dialogue generation, such as in persistent loops or inappropriate content, while otherwise letting the agents improvise autonomously.

4.1.5. Stage V: critique and revision

This stage consists of two steps:

- **Evaluation.** The critic agent assesses both story-level structure (e.g., logical coherence, narrative completeness) and character-level performance (e.g., consistency, emotional alignment).
- **Suggestion.** It offers corrective and creative feedback. Creative suggestions are informed by the knowledge graph, which provides context-aware narrative inspirations for further refinement. The feedback can be used either by the Director for manual refinement or directly by the agents to reinitiate the generation cycle.

The evaluation results include structured feedback and revision suggestions, enabling the Director to decide whether to revise, extend, or conclude the scene. This stage forms a feedback loop that facilitates iterative refinement and quality control.

4.2. Agent design

While the previous section introduced the structural composition of Co-DIRECT, we now focus on the internal design and operational logic of the three agents. Each agent executes a specialized subtask, including narrative planning, character improvisation, or quality assessment, based on its unique configuration and access to the shared knowledge graph. By formalizing agent behaviors through structured procedures and leveraging narrative-grounded knowledge, the system ensures interpretability and creative depth in drama generation.

4.2.1. Writer agent

The Writer Agent accepts high-level narrative settings as input and generates structured character biographies and narrative arcs. It first queries the knowledge graph to retrieve reference character archetypes and associated narrative tags. These tags, reflecting key emotional or event-driven inspirations, are used to synthesize detailed biographies through iterative expansion.

For narrative planning, the agent supports two alternative plot structuring strategies: the linear-sequence-based five-act structure and the

character-centered Hero's Journey structure. An initial narrative arc is generated based on the director's configuration, and then iteratively enriched using creative tags retrieved from the knowledge graph. Structural refinements are subsequently applied to ensure narrative coherence, dramatic pacing, and creative richness (Algorithm 1).

Algorithm 1 Writer agent procedure.

Require: Settings, KnowledgeGraph

Ensure: Biographies, NarrativeArc

```

1:  $Tags \leftarrow QueryKG(Settings)$ 
2:  $Bios \leftarrow GenerateBiographies(Settings, Tags)$ 
3:  $Arc_0 \leftarrow GenerateInitialArc(Settings, Bios)$ 
4:  $Arc \leftarrow EnrichArc(Arc_0, Tags)$ 
5: return  $Bios, Arc$ 

```

4.2.2. Actor agent

The Actor Agent operates on the generated narrative arc and character biographies to simulate dialogue within each scene. For every scene, it formulates performance goals that define the scene's target state.

Dialogue is generated turn-by-turn. The Actor Agent sequentially embodies each speaker, producing the next utterance from scene context and prior dialogue. It dynamically infers the next speaker from history and cues, enhancing diversity and realism. Each turn conditions on preceding dialogue and the character biography until the performance goal is met (Algorithm 2).

Algorithm 2 Actor agent procedure.

Require: NarrativeArc, Biographies

Ensure: SceneDialogues

```

1:  $Dialogues \leftarrow []$ 
2: for  $scene$  in  $Arc$  do:
3:    $DialogHistory \leftarrow []$ 
4:    $Goal \leftarrow GenerateSceneGoal(scene)$ 
5:    $Context \leftarrow InitializeSceneContext(scene)$ 
6:   while  $\neg GoalReached(DialogHistory, Goal)$  do:
7:      $spk \leftarrow NextSpeaker(DialogHistory, Context)$ 
8:      $Utterance \leftarrow GenUtter(spk, DialogHistory, Bios)$ 
9:      $DialogHistory.append(Utterance)$ 
10:     $Context \leftarrow UpdateCtx(Context, Utterance)$ 
11:   end while
12:    $Dialogues.append(DialogHistory)$ 
13: end for
14: return  $Dialogues$ 

```

4.2.3. Critic agent

The Critic Agent provides post-generation assessment and feedback for both narrative structure and character interactions. It aligns all evaluations with the high-level configurations provided by the human director, which serve as reference anchors for dramatic tone, character identity, and thematic arc.

To assess structural coherence, the agent cross-checks the generated narrative arc against the director's configuration, identifying any contradictions, logical breaks, or misalignments with intended story progression. It also detects redundant content, defined as unintentional reiteration of previously established information, such as repeated scene actions, redundant dialogue lines, or stagnant emotional beats. Similarly, it evaluates character dialogues for consistency with character biographies and dramatic goals, focusing on whether each character speaks and acts in accordance with their defined traits.

We implement a dual-evaluation mechanism involving two coordinated agents:

- **Red Critic.** Identifies potential issues such as contradictions, broken causal links, or redundant content in the narrative and dialogue.

- **Blue Critic.** Reviews each flagged issue, offering either supportive elaboration or reasoned counterarguments to determine whether the issue constitutes a genuine narrative flaw.

This adversarial exchange enables fine-grained detection and contextual interpretation of problems, including personality inconsistencies, disrupted logic, thematic stagnation, or dialogue repetition (Algorithm 3).

Algorithm 3 Dual critic agent.

Require: NarrativeArc, SceneDialogues, Settings

Ensure: ValidatedIssues, CounterArguments, Suggestions

```

1:  $VI \leftarrow RedCritic(Arc, Dialogues, Settings)$ 
2:  $CA \leftarrow BlueCritic(VI, Arc, Dialogues, Settings)$ 
3:  $Sug \leftarrow GenSuggestions(VI, CA, Arc, Dialogues)$ 
4: return  $VI, CA, Sug$ 

```

In addition to evaluative reasoning, we also incorporate a creative-oriented critic to support novelty and stylistic diversity.

To provide traceable and explainable inspiration feedback, the agent queries the knowledge base to retrieve creative tags. This process is conceptually similar to the tag retrieval mechanism employed by the Writer Agent during character and arc generation (see Section 4.3.1).

For each scene in the narrative arc, the Critic Agent constructs a context-aware query by summarizing all previously generated events and identifying the narrative thematic focus and character performance focus of the current scene. This combination is used to query the knowledge graph and retrieve potentially useful creative tags. Similar to the handling of character dialogue sequences, this adopts a comparable processing routine. The results are then formatted into structured suggestions and returned to the human director (Algorithm 4).

Algorithm 4 Creative critic agent.

Require: KnowledgeGraph, Setting, NarrativeArc, SceneDialogues

Ensure: CreativeTags, CreativeSuggestions

```

1:  $CT \leftarrow []$ 
2:  $CS \leftarrow []$ 
3:  $PreScene \leftarrow []$ 
4: for  $scene$  in  $Arc$  do:
5:    $S_{PreScene} \leftarrow SummarizePreScenes(PreScene)$ 
6:    $Focus \leftarrow IdentifySceneFocus(scene)$ 
7:    $Query \leftarrow ComposeQuery(S_{PreScene}, Focus)$ 
8:    $Tags \leftarrow QueryKG(Query)$ 
9:    $Suggestion \leftarrow FormatSuggestion(scene, Tags)$ 
10:   $CT.append(Tags)$ 
11:   $CS.append(Suggestion)$ 
12:   $PreScene.append(scene)$ 
13: end for
14: return  $CT, CS$ 

```

4.3. Implementation details

We construct the ontology via automated prompt engineering. Essential a-priori knowledge distilled from classical narratological treatises is formalized into extraction rules, which are instantiated as prompt templates to guide a scripted batch preprocessor in identifying key dramatic elements from the raw corpus. The prompt templates used in this stage are provided in Appendix B. The extracted triples are organized into an ontology-driven knowledge graph, with named entity types including *person*, *organization*, *scene*, *location*, *time*, *key.item*, *key.clue*, *key.event*, *person.characteristic*, and *item.characteristic*. Relations in the graph capture semantic and narrative connections among these entities. We utilize Neo4j for efficient graph storage and querying, and incorporate GraphRAG (Edge et al., 2024) to improve retrieval accuracy and

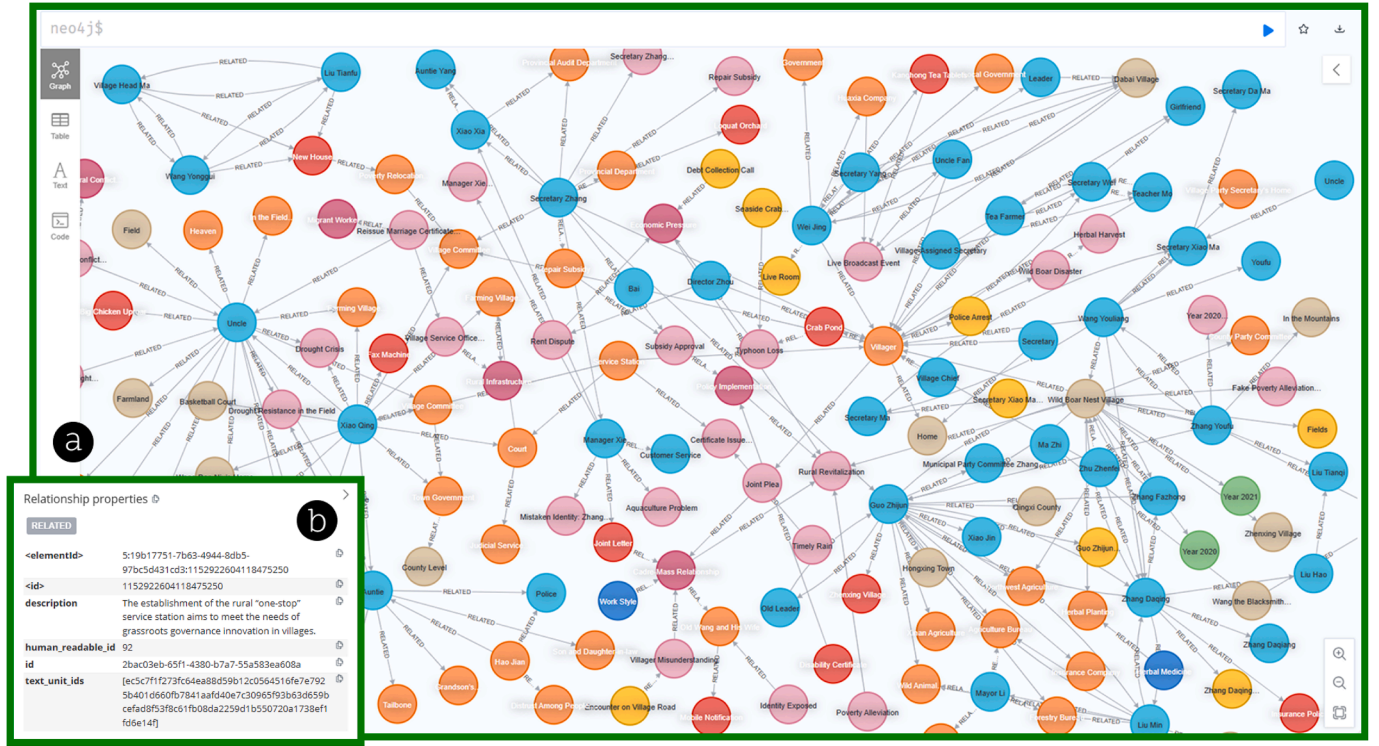


Fig. 4. A visualization of the constructed knowledge graph in Neo4j. In the main window (a), nodes represent entities (e.g., characters, locations, events), while edges capture narratological relations extracted from drama scripts. The specific properties of each relationship are displayed explicitly in the secondary window (b), including the relationship's ID and a detailed description.

knowledge-grounded generation. The constructed graphs in our corpus (48 lengthy scripts) contain on average 4162 entities and 4684 relations, indicating large-scale processing. Importantly, GraphRAG naturally supports incremental corpus expansion, allowing the knowledge graph to scale without full reconstruction when new scripts are added.

To demonstrate the structure and complexity of the constructed knowledge graph, we present a visualization of an actual instance built in Neo4j. As shown in Fig. 4, the graph connects various narratological entities through semantically meaningful relations extracted from drama scripts. This visual representation exemplifies how abstract narrative concepts are operationalized in the implementation layer, supporting semantic retrieval, creative planning, and agent-level reasoning during generation.

The drama script generation pipeline is powered by large language models, with the entire Co-DIRECT framework deployed locally to ensure low latency and direct accessibility. Specifically, Qwen2.5-72B serves as the foundational LLM for the Writer, Actor, and Critic agents, as well as for knowledge retrieval tasks. The foundational LLM runs on a cluster of four NVIDIA RTX 3090 GPUs. During knowledge retrieval, the LLM temperature is set to 0 for deterministic responses, while for creative generation, it is set to 0.7 to encourage diversity and originality. We adopt nomic-embed-text as the embedding model to compute vector representations of input texts; it is currently one of the most widely used embedding models in the community for semantic matching within the knowledge base. Additionally, the human-AI interaction interface is implemented using a customized ComfyUI workflow, enabling visualization and flexible intervention by the human director.

We release both the constructed knowledge base and scalability-oriented implementation details to facilitate reproducibility and future extension. All implementation code, prompt templates, and processed knowledge base files are released at <https://github.com/Y-boy/Co-DIRECT.git>.

5. Experiments

5.1. Baseline

The most direct method for script generation applies a general-purpose large language model (LLM) to generate the entire script in a single step. It includes both the overarching storyline and the corresponding dialogues. This *All-at-Once* approach offers convenience and requires minimal human intervention, serves as the simplest baseline, representing the most basic single-step generation strategy outside of the structured paradigms commonly explored in recent SOTA.

Prior research has explored two major strategies aimed at improving narrative coherence and enriching content. One such strategy is the progressive generation paradigm, as exemplified by systems like AI Dungeon. In this approach, the script is generated iteratively, with each new segment building upon the previous content. Due to the token-length constraints of LLMs, earlier parts of the script are often summarized to serve as contextual input for future generation steps. Human input is frequently incorporated to guide the process, ensuring alignment with evolving storylines and enhancing character consistency. This paradigm not only reflects a key SOTA trend in chain-of-thought narrative planning, but also provides an abstract representative of progressive generation methods for fair baseline comparison.

Another approach is the hierarchical generation paradigm, exemplified by systems such as Co-Writing. This method decomposes the script generation pipeline into semantically meaningful stages. Beginning with a logline or high-level premise, the model incrementally generates intermediate elements, including character profiles, plot structures, and scene-level content. This structured pipeline facilitates global planning and enables constraint propagation, thereby maintaining consistency in character motivations, narrative pacing, and thematic development. Furthermore, the approach closely reflects human screenwriting

workflows by providing interpretable intermediate outputs, which enhance both transparency and controllability. As such, hierarchical generation serves as an abstract representative of the layered, structured SOTA methods in script generation, covering systems like Plan-and-Write and other stage-wise LLM-based approaches.

Despite these advances, existing methods often suffer from content homogenization, producing scripts with similar structures, character dynamics, and thematic patterns across different runs under comparable narrative settings. Such uniformity limits the creative utility of these systems, particularly in collaborative or exploratory writing scenarios where diversity of output is crucial.

Therefore, we select three representative baselines: All-at-Once, Progressive, and Hierarchical. These approaches embody the dominant paradigms of contemporary long-form script generation. This abstraction allows us to provide a fair and informative comparison, even without directly replicating every recent SOTA implementation. It is worth noting that recent state-of-the-art (SOTA) systems for long-form drama script generation, such as Co-Writing, chain-of-thought narrative planners, and plan-and-write pipelines, are not reproduced individually in our experiments due to implementation complexity and reproducibility challenges. However, as discussed in Section 2, these systems can be abstractly categorized into the progressive and hierarchical paradigms. By selecting these paradigms as representative baselines, alongside the simplest All-at-Once strategy, our evaluation setup ensures coverage of the dominant methodological families in DSG. This abstraction allows us to fairly benchmark against the underlying principles of current SOTA approaches, while maintaining experimental feasibility and comparability.

In the following section, we evaluate three representative baseline strategies for script generation: (1) the All-at-Once approach, (2) the progressive generation approach, and (3) the hierarchical generation approach. The evaluation considers three key aspects: (i) the overall length of the narrative structure and dialogues, (ii) the degree of content de-homogenization across multiple generations, and (iii) the coherence of storylines across scenes.

5.2. Experimental setup

To evaluate different script generation paradigms under consistent creative constraints, we designed a controlled experiment comparing our proposed multi-agent method with three representative baselines. The entire setup is grounded in a real-world co-creation scenario that simulates how AI systems assist human creators in producing long-form narrative scripts.

5.2.1. Human-AI co-creation context

An experienced human director was invited to initiate and participate in the script co-creation process. The invited directors provided a high-level creative specification encompassing cultural, historical, and geographical backgrounds, core character descriptions, emotional baselines, and the intended target audience (see Appendix A). All methods received the same input specification, including our proposed framework and the three baseline strategies described in Section 5.1.

In this experiment, all human-AI co-creation processes focused on the shared theme of the revitalization of the hometown, involving a storyline where the protagonist collaborates with villagers to collectively rebuild their rural homeland. We first collected a wide range of scripts from publicly available online communities and platforms (e.g., Baidu Wenku, Zhihu Columns). After deduplication, the invited directors then reviewed and filtered this collection, resulting in 48 classical scripts that align with the target theme. These scripts are widely recognized in theater studies as representative exemplars of narrative and structural diversity, covering genres such as historical drama, sketch comedy, or straight play (see Appendix C for the full list). While this corpus provides a solid foundation of narratological knowledge, it is sourced from open

Chinese-language communities and therefore primarily reflects the narrative styles and cultural contexts of modern Chinese drama, which may introduce cultural bias. We acknowledge this limitation and plan to extend the corpus with works from a broader range of cultural traditions in future work. All scripts were further processed following the ontology-based methods described in Section 3: they were digitized, segmented into scenes and dialogues, and annotated with metadata such as character roles and thematic motifs. For each method, 20 scripts were independently generated under identical input conditions. The goal is to evaluate whether the systems can consistently produce multiple long-form script candidates that are both narratively coherent and meaningfully diverse.

5.2.2. Evaluation protocol

Recognizing the cognitive burden and inconsistency that often arise in human evaluation of long-form narratives, we adopted a GPT-assisted evaluation protocol. As described in the Experimental Setup section, all AI generation methods employed in this study operate independently of GPT. This ensures that the scoring model does not influence or bias the content generation process. To ensure fairness and interpretability, the human director specified an explicit constraint set, outlining desirable and undesirable script features, and also provided a reference script as a comparative baseline. This reference script was scored at 3 out of 5 across five evaluation dimensions: Narrative Coherence, Plot Progression, Emotional Realism, Thematic Depth, and Originality & Diversity. This reference script, assigned a score of 3 out of 5 across the five evaluation dimensions, serves as a standardized scoring anchor. GPT subsequently applied this anchor-based template to evaluate all generated scripts in a consistent and scalable manner, thereby enabling the assessment of 160 scripts without necessitating manual scoring for each instance. A GPT-based model assigned a score from 1 to 5 to each generated script in each dimension, using the reference script as an anchor. The model was prompted with the constraint set and reference script in each evaluation instance to ensure consistency and context awareness. By employing this reference-based scoring protocol, the evaluation remains reproducible and mitigates concerns regarding circular validation bias, given that GPT is utilized solely for scoring outputs that are generated independently. This evaluation setting was applied uniformly across all baseline and ablation variants to ensure fair and reproducible comparisons.

Statistical significance tests were also conducted as a consistency check. Two-sample *t*-tests between Co-DIRECT and the strongest baseline indicate significant improvements ($p < 0.05$) in Narrative Coherence and Originality & Diversity. These results confirm that the observed gains are systematic rather than random fluctuations.

5.2.3. Automated diversity metrics

Traditional lexical-level similarity metrics often fall short in capturing structural and semantic diversity in complex narrative content. In particular, word-level similarity is not representative for evaluating narrative arcs or character dialogues, which typically exhibit variation at the sentence or paragraph level. Therefore, we adopt embedding-based similarity measures that operate on larger textual units, enabling a more meaningful assessment of diversity across scripts.

To evaluate content diversity more comprehensively, we employed both embedding-based similarity analysis and structural statistics derived from script-level distributions. These metrics capture lexical repetition, dialogue uniformity, and structural convergence across generations.

Average narrative structure similarity. Computes the mean pairwise cosine similarity between narrative arcs of scripts generated by each method. Lower values indicate higher narrative variety.

Average dialogue similarity. Computes the mean pairwise cosine similarity of scene-level dialogue embeddings, assessing how similar the dialogue flows are across scripts.

To quantify average similarity, we define the following:

$$\text{Sim}_M = \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} \text{cosine_similarity}(v_i, v_j)$$

where v_i and v_j denote the embedding vectors of scripts (or dialogue segments) generated by method M , and n is the number of generated scripts.

High-similarity ratio. This metric measures the proportion of dialogue pairs whose cosine similarity exceeds three defined thresholds: 0.85, 0.90, and 0.95. A higher ratio indicates a tendency toward homogenized phrasing or the repeated use of structural templates.

To formalize this metric:

$$\text{Ratio} = \frac{|\{(i, j) \mid 1 \leq i < j \leq m, \text{cosine_sim}(v_i, v_j) \geq \tau\}|}{\frac{m(m-1)}{2}}$$

where $\tau \in [0, 1]$ is the similarity threshold (e.g., 0.85, 0.90, 0.95); the numerator counts the number of dialogue pairs exceeding the threshold, and the denominator is the total number of unique dialogue pairs.

Redundancy and information density. To quantify repeated or overlapping content across all scripts generated by a method, we compute redundancy and information density based on extracted event units. Let all events from M scripts be $E_{\text{all}} = \{e_1, e_2, \dots, e_{N_{\text{tot}}}\}$, where each event e_i has embedding $v_i \in \mathbb{R}^d$ and N_{tot} is the total number of events across all scripts.

The **redundancy** is defined as the average pairwise cosine similarity among all events:

$$R = \frac{2}{N_{\text{tot}}(N_{\text{tot}} - 1)} \sum_{i=1}^{N_{\text{tot}}-1} \sum_{j=i+1}^{N_{\text{tot}}} \frac{v_i \cdot v_j}{\|v_i\| \|v_j\|}$$

The **information density** is then:

$$D = 1 - R$$

To capture variability across scripts, we report the standard deviation of redundancy and density computed on each individual script:

$$\sigma_R = \sqrt{\frac{1}{M} \sum_{k=1}^M (R_k - \bar{R})^2}, \quad \sigma_D = \sqrt{\frac{1}{M} \sum_{k=1}^M (D_k - \bar{D})^2}$$

5.2.4. Ablation studies

To better understand the contribution of individual components within our proposed framework, we conducted a series of ablation studies. The same evaluation protocol and diversity metrics described above were applied to the ablation settings, ensuring comparability between baseline and internal model variants. Specifically, we designed three ablated variants:

- **Without Knowledge Base Preprocessing:** Removes structured knowledge extraction and triple-based indexing, relying instead on unprocessed corpus inputs for contextual grounding.
- **Without Situation Decomposition and Improvisation:** Replaces the actor agent's multi-turn reasoning and improvisational dialogue generation with a single-pass response based solely on the narrative arc.
- **Without Multi-round Evaluation and Revision:** Eliminates the critic agent's feedback loop, disabling iterative refinement between the writer and actor agents.

These ablation settings allow us to isolate the effect of knowledge grounding, interaction modeling, and iterative revision in the overall performance of the multi-agent framework.

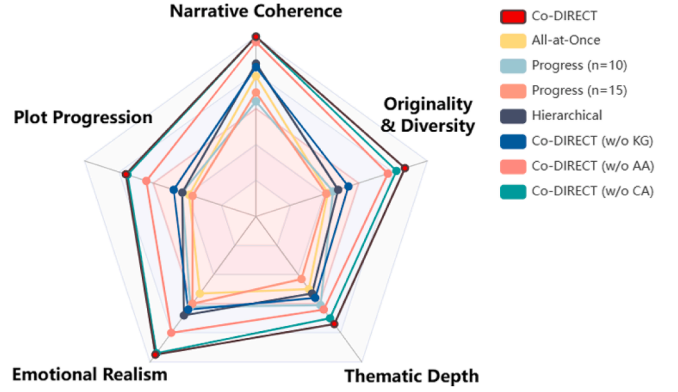


Fig. 5. GPT-based evaluation scores across five content quality dimensions.

5.3. Results and analysis

This section analyzes experimental results for all evaluated methods, including our multi-agent framework and its ablated variants, focusing on: (1) GPT-based content quality, (2) dialogue diversity, and (3) script length and structure.

5.3.1. Content quality assessment

As summarized in Table 1, all AI-driven methods, including baseline systems and ablation variants, achieved scores in Narrative Coherence that met or exceeded the reference script benchmark. The benchmark script was manually rated 3 across all five evaluation dimensions, and all model-generated scripts were evaluated relative to this standard. Our proposed Co-DIRECT framework outperformed all other methods across all dimensions. In particular, it achieved the highest ratings in Narrative Coherence and Plot Progression, which reflects better handling of long-range narrative dependencies and causal relationships between events. It also achieved higher scores in Emotional Realism and Thematic Depth, due to the agent-based architecture's ability to generate more nuanced and layered story developments. The Originality and Diversity dimension exhibited the greatest variance across systems. Baselines such as All-at-Once and lower-step Progressive variants tended to produce formulaic outputs with limited novelty. In contrast, Co-DIRECT maintained a high level of creative variation.

Ablation studies showed that removing either the knowledge graph construction module (w/o KG) or the Actor Agent (w/o AA) led to consistent declines across all dimensions except Narrative Coherence. The performance drop without the knowledge graph indicates that external knowledge plays a critical role in enhancing both the Writer and Critic agents. Meanwhile, the degradation caused by removing the Actor Agent highlights the importance of decomposed shot modeling and improvisational dialogue in contributing to emotional and thematic richness.

For ease of comparison, we also visualize the results in a radar plot (Fig. 5). While the table provides exact numerical scores, the radar chart highlights relative strengths and weaknesses across evaluation aspects.

5.3.2. Diversity evaluation

To assess diversity across both narrative structure and linguistic content, we employed three metrics: Average Narrative Structure Similarity, Average Dialogue Similarity, and High-Similarity Ratio. These metrics collectively quantify how similar the generated outputs are in terms of plot development and dialogue patterns.

As illustrated in Fig. 6, our Co-DIRECT framework achieved the lowest average similarity scores in both narrative structure and dialogue content, indicating the greatest diversity across its 20 generated scripts. In contrast, All-at-Once and Progressive methods exhibited average similarities exceeding 0.8 in both dimensions, signaling a pronounced tendency toward template-based generation and reduced content variety.

Table 1

Quantitative evaluation scores of different DSG approaches. The scores in the table are the average scores of 20 generated scripts.

Model	Narrative coherence	Plot progression	Emotional realism	Thematic depth	Originality & diversity
Reference Script	3.00	3.00	3.00	3.00	3.00
All-at-Once	3.90	1.95	2.65	2.50	2.10
Progress ($n = 10$)	3.20	2.15	3.10	3.05	2.25
Progress ($n = 15$)	3.45	1.85	3.00	2.15	2.05
Hierarchical	4.25	2.15	3.40	2.65	2.40
Co-DIRECT (w/o KG)	4.15	2.40	3.20	2.80	2.70
Co-DIRECT (w/o AA)	4.85	3.20	4.00	3.20	3.85
Co-DIRECT (w/o CA)	5.00	3.75	4.70	3.50	4.10
Co-DIRECT	5.00	3.80	4.75	3.70	4.35

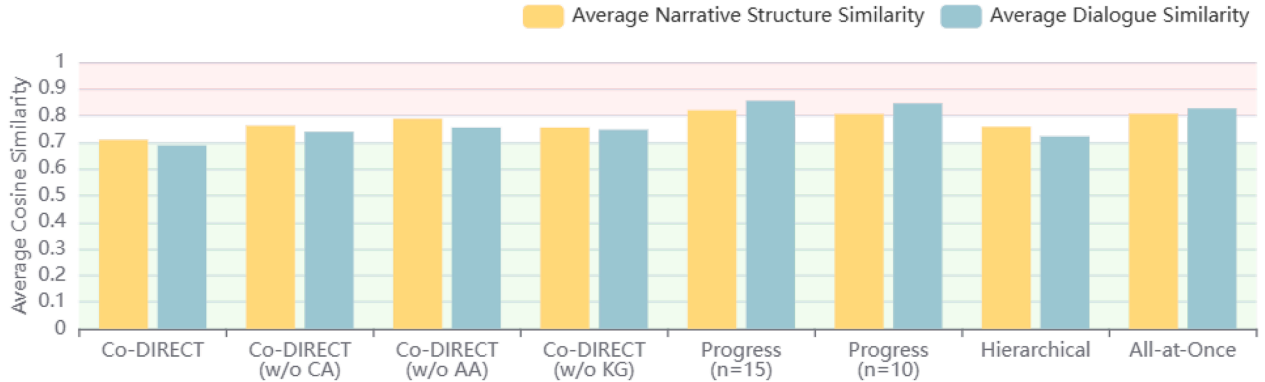


Fig. 6. Average narrative and dialogue similarity across methods. Values above 0.8 indicate strong structural or linguistic overlap across scripts.

High-Similarity Ratio analysis further corroborates these findings (shown as Fig. 7). When the cosine similarity threshold is set at 0.85 or above, the compared dialogues are semantically close but not identical, indicating mild redundancy. At this level, All-at-Once and Progressive methods display the highest proportion of such moderately similar pairs, suggesting early signs of templated outputs. As the threshold increases to 0.90 or above, where dialogue pairs are highly overlapping in meaning, these baseline methods remain dominant in repetition frequency. Finally, at the 0.95 threshold and beyond, where dialogues are nearly identical except for minor differences in phrasing, template reuse becomes most evident. Co-DIRECT consistently maintains minimal overlap across all thresholds. Among ablation variants, disabling actor agents (w/o AA) results in a notable increase in high-similarity pairs, confirming the importance of multi-turn improvisation. The variants without critic agents (w/o CA) or knowledge grounding (w/o KG) show mild increases, reflecting their partial role in maintaining dialogue diversity.

Together, these results confirm that Co-DIRECT effectively mitigates both plot and language-level redundancy through its coordinated multi-agent process, yielding more original and varied script outputs.

5.3.3. Global structure analysis

To analyze the structure and content characteristics of the generated scripts, we examined three indicators: average number of scenes per script, average dialogues per scene, and the average total token count per script. These reflect each method's capacity to produce extended narratives with structurally rich and dialogue-dense content.

As shown in Fig. 8, our Co-DIRECT framework yields the most structurally elaborate outputs, characterized by a balanced combination of high scene counts, dense dialogue exchanges, and extended script lengths. In contrast, the All-at-Once and Hierarchical baselines produce shorter, less developed scripts with limited narrative expansion. Pro-

Table 2

Information density and redundancy of different methods.

Method	Density	Redundancy
All-at-Once	0.3763 $\pm$ 0.0406	0.6237 $\pm$ 0.0406
Co-DIRECT	0.3346 $\pm$ 0.0342	0.6654 $\pm$ 0.0342
Hierarchical	0.3825 $\pm$ 0.0188	0.6175 $\pm$ 0.0188
Progressive ($n = 10$)	0.3466 $\pm$ 0.0364	0.6534 $\pm$ 0.0364
Progressive ($n = 15$)	0.3391 $\pm$ 0.0512	0.6609 $\pm$ 0.0512

gressive methods demonstrate moderate improvement as the number of generation steps increases.

Among the ablation settings, disabling the actor agents (w/o AA) had the most visible impact, leading to significantly lower dialogue density as the system reverted to more generic one-pass generation. The other two ablations, which involved removing knowledge graph construction (w/o KG) and disabling critic agents (w/o CA), produced scripts with overall structure and content length comparable to Co-DIRECT, but exhibited subtle reductions in balance and richness. These results suggest that while Co-DIRECT, w/o KG, and w/o CA all maintain a similar narrative scale, only the full model consistently achieves fine-grained structural elaboration.

These findings demonstrate the necessity of multi-agent coordination and structured knowledge integration in producing coherent, large-scale scripts.

5.3.4. Cross-script redundancy analysis

As shown in Table 2, our experiments reveal several insights regarding redundancy and information density across generated scripts.

First, for the Progressive methods with different iteration lengths, increasing the number of iterations from 10 to 15 results in an upward trend in redundancy, indicating that longer generation length can impact both redundancy and information density.

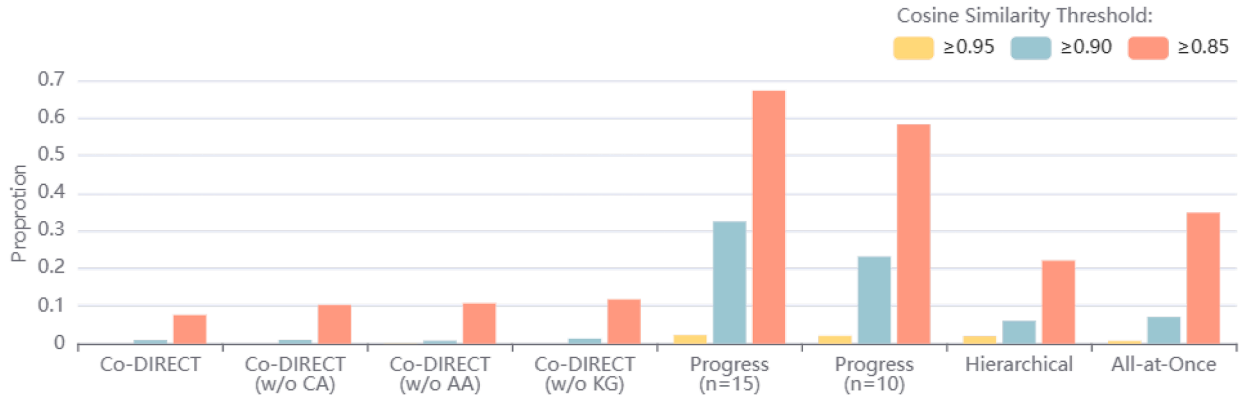


Fig. 7. Distribution of high-similarity dialogue pairs at thresholds of 0.85, 0.90, and 0.95. Higher bars indicate greater redundancy.

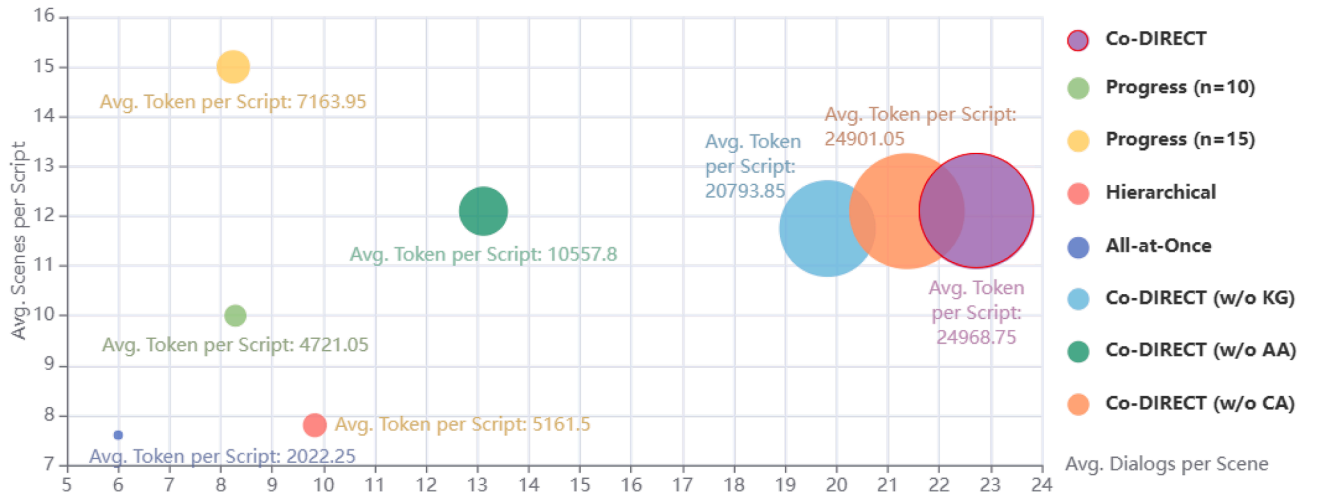


Fig. 8. Structural characteristics of generated scripts across methods. The X and Y axes represent average dialogues per scene and average scenes per script, respectively. Circle size indicates the average total token count per script.

Second, although Co-DIRECT generates substantially longer scripts compared to the baselines, the redundancy and information density metrics remain comparable, demonstrating that our method can generate richer content while maintaining undiluted information density. Nevertheless, the extent to which the size of our knowledge base may influence the information density of our approach remains an interesting avenue for future exploration.

Finally, the standard deviation analysis shows that the Hierarchical method exhibits the smallest variability across scripts, indicating the highest level of homogenization. This is consistent with the top-down global constraints imposed by the method, which limit diversity across generated scripts.

5.3.5. Summary of findings

In summary, Co-DIRECT demonstrates consistent advantages across structural scale, content quality, and diversity metrics. It produces longer, well-structured scripts with higher GPT-assessed quality and substantially reduced redundancy in both narrative and dialogue levels. The ablation results affirm that knowledge grounding, actor improvisation, and critic-driven revision each play a vital role in achieving these outcomes.

In the next section, we present a qualitative case study to illustrate how these improvements manifest at the level of narrative and character interaction.

5.4. Case study

5.4.1. Alignment with human intent and creative workflow

This section presents a qualitative analysis of 20 full-length scripts generated by Co-DIRECT. Through selected examples, we illustrate how the system aligns with human guidance, fosters creative diversity, and reveals current limitations.

In terms of output quality, a review of the 20 full-length scripts generated under consistent directorial input revealed high alignment with the narrative intentions originally provided. Storylines adhered to the director's high-level configurations, while character personalities, motivations, and emotional progressions largely conformed to the envisioned roles and arcs. Table 3 presents a comparison between selected configuration settings and the corresponding realizations in the generated scripts, illustrating this alignment.

5.4.2. Creative diversity and novel plot expansion

Beyond alignment, Co-DIRECT demonstrates a notable capacity for creative diversity. Despite receiving identical initial configurations, which include character prompts, setting constraints, and narrative goals, the 20 generated scripts exhibit a wide range of plot developments, emotional trajectories, and character fates. This variation reflects the framework's ability to explore the creative space without sacrificing structural integrity.

Table 4 summarizes representative patterns of narrative resolution diversity observed across a subset of scripts. We highlight several

Table 3
Alignment of narrative arcs with original character and theme settings.

Script	Matched setting type	Matched scene excerpt
1	Character Background: Chen from rural area; Zhiyi guided by university values	“Chen grew up in the countryside and aspired to change his life through education; Zhiyi, raised in the city, was influenced by university coursework to work in rural governance.”
2	Plot Motivation: Career transition + Rural leadership role + Tech project collaboration	“Chen leaves his corporate job...Zhiyi becomes a village official...together they initiate a local environmental improvement initiative; Youth entrepreneurship hub is launched.”
3	Plot Motivation & Personal Values: Social responsibility; intergenerational conflict	“Zhiyi was shaped by her university experience...Chen communicates his intention to return home...his father eventually supports the decision.”
4	Family Dynamics & Trust Building: Intergenerational tension and emotional appeal	“I want to contribute to my hometown.; Zhiyi approaches Chen’s father...and shares her vision for rural change. He remains silent, later quietly supportive.”
5	Plot Motivation: Childhood labor memory + Civic commitment	“Chen recalls working in the fields with his father...Zhiyi formally accepts her village officer role, reaffirming her commitment to community service.”
6	Career Decision & Rural Innovation: Urban-to-rural transition + Community skepticism	“Moved by rural challenges and Zhiyi’s example, Chen quits his job to return; Zhiyi, shaped by academic values, chooses to contribute to local transformation after graduation.”
7	Interpersonal Tension & Determination: Parental resistance and reaffirmed conviction	“Chen’s father initially opposes his and Zhiyi’s rural plan...Chen reaffirms his commitment: ‘This is not escape—it’s responsibility.’”
8	Plot Trigger: Personal calling + educational influence	“Chen receives a letter from his hometown, which deeply moves him; Zhiyi had long contemplated grassroots service after being inspired at university.”
9	External Call & Parental Transition: Community invitation and family dynamics	“Both receive invitations encouraging skilled youth to return; Chen’s father gradually expresses cautious support after seeing their dedication.”
10	Career Shift & Institutional Support: Urban resignation + rural leadership + mentorship	“Chen resigns from his urban job; Zhiyi decides to stay in the village; both receive training and guidance from local mentors to support their goals.”
11	Cross-background protagonists return home in response to rural development calling	“Chen decides to resign from his foreign company job after deep internal struggle; Zhiyi, inspired by university teachings, becomes a village official. They meet during a village meeting and decide to collaborate.”
12	Character arc: personal history and rural motivation intersect	“Chen reflects on his childhood labor in the fields and his father’s words: ‘We farmers must do something for this land.’ Zhiyi recalls reading rural development books in the university library and reconnects with Chen to share her aspirations.”
13	Community planning and rural revitalization strategy deployment	“Chen and Zhiyi present a revitalization plan using visual aids to village representatives. Villagers show initial hesitation but are intrigued by the idea of local transformation.”
14	Intergenerational resistance to return-home decisions	“Chen expresses his intention to return to the village during dinner. His father responds firmly: ‘You’re doing fine in the city — why come back and suffer?’”
15	Inspiration from elder community members and rural memory	“A respected village elder shows Chen and Zhiyi a book documenting local history, encouraging them to blend tradition with modern methods for the village’s future.”
16	Youth and art-led ecological innovation	“Together with young villagers and artists, Chen and Zhiyi launch an ecological project integrating murals and reforestation, despite initial conservative resistance.”
17	Emotional encouragement and collective purpose	“Zhiyi reassures a hesitant Chen during a walk: ‘We must not only change ourselves but also this land.’ They resolve to build the future together.”
18	Innovation in rural commerce through digital means	“To promote village products, Chen and Zhiyi try livestream selling. Though technical difficulties arise, with villagers’ help they successfully launch the channel.”
19	Cultural revitalization through arts education	“An aspiring teacher proposes starting an art class. With Chen and Zhiyi’s support, and despite opposition from some elders, they begin a mural project in the village.”
20	Crisis leadership and community resilience	“During a drought, Chen and Zhiyi coordinate a water access response by organizing excavation efforts and emergency supply transfer from neighboring villages.”

illustrative examples directly within the table to showcase how narrative structure, emotional tone, and dramatic outcomes vary across different generated scripts.

5.4.3. System limitations and areas for improvement

Some limitations remain in Co-DIRECT’s current design. The system tends to overemphasize dialogue, often neglecting action cues, emotional nuance. These issues occasionally result in rigid expressions, weak subtext, or repetitive exchanges.

Table 5 presents selected examples highlighting these shortcomings, which point to future directions such as improved multimodal grounding and better agent coordination.

6. Discussion and future work

Our current approach assumes a linear narrative arc, limiting applicability to non-linear or experimental scripts. LLMs struggle to maintain branching threads over long contexts, while existing corpora and metrics remain biased toward linear traditions. Prior work has thus focused on analytic aids (e.g., graph-based models [Chen, 2025](#), visualization [Kim et al., 2017](#)) rather than end-to-end generation. A natural extension is to represent arcs as graph or hypergraph structures with external memory

to support parallel threads. Since non-linear storytelling is inherently subjective, future evaluation must consider both structural coherence and audience experience.

The system tends to overemphasize dialogue while underrepresenting non-verbal narrative elements, such as action descriptions, micro-expressions, and camera cues, which are critical to cinematic realism and performative depth. Second, while the multi-agent architecture promotes diverse output, it occasionally yields inconsistencies in emotional flow or scene transitions, revealing a need for tighter coordination among agents. Third, the current framework is limited to text-based generation; incorporating multimodal elements such as gesture, prosody, or visual cues could significantly enhance script expressiveness. Future work will focus on finer-grained emotional modeling, stronger agent coordination, and multimodal extensions toward embodied storytelling.

A limitation of our current experiments is that they were conducted within director-specified genres, without systematic cross-genre validation. Co-DIRECT’s design, based on core narrative abstractions (characters, conflicts, causal links), is theoretically extensible to diverse genres (e.g., tragedy, comedy). Future work will explore genre-aware prompts, corpora, and evaluation metrics to test this potential.

Table 4
Innovative elements identified in generated scripts.

Thematic category	Script	Innovation description	Matched scene excerpt
Organizational Mechanisms	2	Establishment of a youth incubator platform for rural entrepreneurship	"They launched a 'Rural Youth Innovation Incubator' to attract more young people to return and start businesses."
	9	Use of formal invitations by the village committee to encourage return	"They received formal invitations from the village committee."
	13	Multimedia-supported planning presentation to villagers	"They presented their plan using projections and visuals to engage villagers."
Ecology and Aesthetic Integration	16	Rural art + child participation + tree planting for ecological renewal	"Murals were painted, children joined the art process, and villagers planted trees together."
	19	Launch of children's art class with community mural painting	"An art class was launched for rural children, transforming village walls through collaborative murals."
Conflict and Realism	6	Skepticism from villagers about new technology projects	"Some villagers were skeptical of the project, fearing high risks."
	18	Technical failure in livestream commerce, resolved by villagers	"The first livestream failed due to technical issues; a villager helped fix the equipment."
	20	Drought crisis prompts coordinated water rescue by youth	"They led villagers to dig a new water source and transport emergency supplies from nearby communities."
Return Motivation Diversity	8	Personal letter from hometown sparks emotional return	"He received a letter from home, which deeply moved him."
	12	University reading on rural topics motivates self-driven return	"She read a rural development casebook in college and felt inspired to act."
	14	Dining-table conflict reveals realistic parental disapproval	"You're doing well in the city—why come back to suffer?"
Intergenerational Tension & Repair	4, 7	Father gradually accepts son's return via silent observation	"Chen's father watched them work in silence, slowly coming to terms."
	15	Village elder provides historical continuity and moral encouragement	"An elder showed them an old village chronicle and encouraged blending past with present."
Professional Identity & Collaboration	5	Female protagonist publicly accepts leadership via appointment document	"Zhiyi showed her official appointment letter, firmly stating her commitment."
	11	Local leader actively proposes community project strategy	"A local leader organized a village meeting and proposed concrete action plans."
	17	Zhiyi provides emotional leadership and shared vision	"We must change ourselves—and also this land."

Table 5
Case samples demonstrating narrative gaps and lack of cinematic detail.

Description	Matched scene excerpt
The action descriptions, micro-expressions, and camera language are not rich enough.	...Villager A(with a broad smile): Zhiyi, hearing you say this makes us feel much more at ease. ...Zhiyi(smiling): Thank you for your support!
The content is pale and rigid.	..."Chen submitted his resignation to the company. Though he felt a tinge of reluctance, the thought of contributing to his hometown filled him with strength.""At the village assembly, Chen introduced an environmental improvement project, which was met with concern and opposition from some villagers. The next morning, the two discussed the situation and decided to win support through a pilot demonstration." ...

7. Conclusion

This paper presented **Co-DIRECT**, a multi-agent framework for human-guided drama script generation. By simulating professional screenwriting workflows and embedding a *Director-in-the-Loop (DiL)* paradigm, Co-DIRECT integrates Writer, Actor, and Critic Agents with an ontology-driven knowledge graph to support narrative coherence, expressive character development, and creative diversity. Experiments and case studies show that the framework aligns closely with human intent while expanding beyond rigid template-based generation. Co-DIRECT provides a generalizable foundation for controllable, narratively grounded, and human-AI collaborative storytelling, with potential extensions to multimodal performance, interactive co-authoring, and broader applications in computational creativity.

CRediT authorship contribution statement

Boai Yang: Conceptualization, Methodology, Writing – original draft, Validation; **Jiapai Peng:** Software, Validation, Writing – original draft; **Shiyi Tang:** Software, Validation, Writing – original draft; **Jiani Tan:** Software, Validation; **Fengbo Zhou:** Software, Validation; **Shenglan Cui:** Writing – review & editing; **Wei Qu:** Validation; **Tao Li:** Conceptualization, Funding acquisition, Writing – review & editing; **Fang Liu:** Conceptualization, Writing – review & editing; **Dong Wang:** Methodology.

Ethics policy

In this study, publicly available drama scripts from Baidu Wenku were used for non-commercial research, ensuring compliance with copyright and licensing agreements. All characters in the generated scripts are fictional, and we do not condone the use of AI to generate immoral content.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Script background and settings

This section introduces the background of a co-created script, including character relationships, complete setting design, and plot flow. Chen is a young man who resigned from a foreign enterprise to return to his hometown to engage in rural revitalization; Zhiyi is a village official from the city, determined to lead the villagers to prosperity; and Chen's father is a typical Chinese farmer, who is both a father and the village head, with a complex inner world but always committed to his hometown. Through the choices and struggles of Chen and Zhiyi, the script showcases the active participation and selfless dedication of contemporary youth in rural revitalization, conveying the theme that personal dreams are closely linked to the fate of the nation. The film emphasizes the growth and awakening of the younger generation in response to the call of the times. They are not afraid of difficulties, dare to innovate, and use technology and wisdom to assist rural revitalization, demonstrating the vitality and creativity of the youth. All character archetypes are derived from classic characters presented in our knowledge base. All generated characters are fictional to prevent LLM from directly recognizing the character it is role-playing. Table 6 present the drama script settings given by human directors.

Table 6
Script settings in the co-creation.

Script Name: "Youth on the Ridge"
Character Settings: <Character 1> ## Casting Needs: Resilient, Optimistic, Positive ## Requirements for Fictional Character Design: - The character is Chen, a male with a simple and honest appearance, slightly overweight, and raised in a rural area. He studies hard, hoping to get into a university and leave the countryside to strive for success in a big city. Later, he joins a foreign company, but for various reasons, he is inspired by the rural revitalization efforts in his home country and the contributions of Zhiyi to their hometown. - Resilient, optimistic, and positive. - As the protagonist, he drives the progression of the main storyline. <Character 2> ## Casting Needs: Cheerful, Enthusiastic, Kind ## Requirements for Fictional Character Design: - The character is Zhiyi, a young woman who is beautiful, fashionable, and outgoing. She grew up in the city and has never been to the countryside. Influenced by her university's ideological education, she decided to become a village official after graduation, aiming to lead the villagers to prosperity. - Cheerful, Enthusiastic, and kind. - As a supporting character, she drives the development of the subplot. <Character 3> ## Casting Needs: A typical image of a Chinese farmer ## Requirements for Fictional Character Design: - The character is Chen's father, a man with a simple and humble appearance. He is dedicated to leading his hometown to prosperity and hopes his son, Chen, can achieve success in the big city. However, he struggles to accept his son's decision to return home and work in agriculture, leading to complex emotions. - A typical image of a Chinese farmer. - He runs through both the main plot and the subplot, serving as a bridge between the protagonist and supporting characters. # Plot Outline Configuration: ## Basic information of script: - Script Genre: Mainstream-themed film or drama. - Target Audience: Youth group. ## Thematic tone of script: - Emotional tone: Inspirational, youthful, and full of energy. - Core idea: Rural revitalization, responding to the government's call, and building the hometown. - Story background: Against the backdrop of rural revitalization, everyone responds to the government's call and returns to their hometowns to contribute to its development. ## Storyline: The movie Youth on the Ridge profoundly showcases the active participation and selfless dedication of contemporary youth in rural revitalization. Through the choices and struggles of the protagonists, Chen and Zhiyi, the film conveys the deep affection of the younger generation for their hometowns and their positive response to national policies. The film deeply reveals the theme of how personal dreams are closely linked to the nation's destiny. Chen gives up a high-paying job in a foreign company and resolutely returns to his hometown to engage in rural revitalization, while Zhiyi, a city girl, boldly chooses to work as a village official in the countryside, leading the villagers to prosperity. Their youth blossoms in the vast fields, with their actions embodying responsibility and commitment. Their transformation reflects the growth and awakening of youth in response to the call of the times. Undeterred by difficulties, they bravely innovate, using technology and wisdom to contribute to rural revitalization, demonstrating the vitality and creativity of young people. The film promotes positive values, encouraging more youth to devote themselves to rural revitalization and contribute their youthful strength to the realization of the Chinese Dream of national rejuvenation. At the same time, the film also reflects the hardships and achievements of rural revitalization, as well as the support and guidance of national policies for youth entrepreneurship.

Appendix B. Prompts used in Co-DIRECT

We provide the details of the prompt templates. In the knowledge deconstruction stage, the prompt templates in Tables 7–10 are employed to extract both background and comprehensive tags. The Writer agent uses the prompt templates in Tables 11–13 to perform tasks such as retrieving creative tags from the knowledge base, designing fictional characters, and generating narrative arcs. The Actor agent executes improvisational performance tasks using the prompts in Table 14, generating in-character dialogue content. The Critic agent evaluates the script using the prompts in Table 15, and retrieves further usable creative tags and suggestions from the knowledge base via Table 16.

Table 7
Prompt for deconstruction script.

Extracting Background Labels
Task 1 Please read the following script: """ {script} """ # Task 2 From the script, identify the following types of information: *characters*, *items*, *scenes*, *events*, and *issues*. Extract their descriptions and organize them into 'label-description' pairs. **Notes** 1. Extract as many labels as possible from the five categories [character, item, scene, event, issue]. 2. Provide descriptions from multiple dimensions whenever possible. 3. For characters, the description must include family relations, social relations, and personality traits. 4. For items, the description must include their symbolic meaning. # Task 3 Finally, output only the formatted 'label-description' pairs strictly in the following json format: **Notes** 1. Ensure that the output strictly follows the json format requirements: do not omit quotation marks. 2. Ensure that each label, description, as well as names for characters, items (ITEM_NAME), events (EVENT_NAME), issues (ISSUE_NAME), and scenes (SCENE_NAME) are all enclosed in double quotes. # Output Example '''json { "character": ["label": "CHARACTER_NAME", "description": "CHARACTER_DESCRIPTION", ...], "item": ["label": "ITEM_NAME", "description": "ITEM_DESCRIPTION", ...], "scene": ["label": "SCENE_NAME", "description": "SCENE_DESCRIPTION", ...], "event": ["label": "EVENT_NAME", "description": "EVENT_DESCRIPTION", ...], "issue": ["label": "ISSUE_NAME", "description": "ISSUE_DESCRIPTION", ...] } '''

Table 8
Prompt for extracting character profiles.

Character Profile Extraction
<pre># Task 1 Please read the following 'script', 'character', and 'event': ## Script {script} ## Characters {character} ## Events {event} # Task 2: Based on the 'character' tags, 'event' tags, and 'script' content, generate the background, important experience events, and behavioral motivations for each appearing character, and form a standardized character profile. # Task 3: Finally, output only the formatted 'character profiles' strictly in the following json format: # Output Example ''' json { "CHARACTER_NAME": ["background": "CHARACTER_BACKGROUND", "event": "CHARACTER_EVENT", "motivation": "CHARACTER_MOTIVATION"], "CHARACTER_NAME": ["background": "CHARACTER_BACKGROUND", "event": "CHARACTER_EVENT", "motivation": "CHARACTER_MOTIVATION"] ... } '''</pre>

Table 9
Prompt for extracting narrative arc.

Narrative arc extraction
<pre># Task 1 Please read the following 'script', the basic 'narrative_arc_structure', the 'label_dictionary', and the 'character_biography': ## Script {script} ## Nar_arc_structure {narrative arc structure} ## Label_dictionary {label dictionary} ## Character_biography {character biography} # Task 2 Determine the narrative arc type of the script. **Notes** If the arc is a composite type, specify clearly which narrative arc structures are combined. # Task 3 Based on the identified narrative arc type from Task 2, output the formatted 'narrative arc' strictly in the following json format: ''' json { "nar_arc_structure": "NAR_ARC_STRUCTURE", "narrative_arc": "NARRACTIVE_ARC" } '''</pre>

Table 10

Prompt for analyzing comprehensive information.

Comprehensive script information analysis
<pre> # Task 1 Please read the following 'script', the character biographies ('cha_biography'), and the narrative source ('narrative_src'): ## Script {script} ## Cha_biography {cha_biography} ## Narrative_src {narrative_src} # Task 2 Based on the script, the character biographies, and the narrative source, perform a comprehensive, in-depth analysis of the entire script from three perspectives: Language_Style, Narrative_Logic, and Emotional_Tone. # Task 3 Finally, output only the formatted "script critique" strictly in the following json format: ' 'json { "Language Style": ["Language Style1", "Language Style2", "Language Style3", ...], "Narrative Logic": ["Narrative Logic1", "Narrative Logic2", "Narrative Logic3", ...], "Emotional Tone": ["Emotional Tone1", "Emotional Tone2", "Emotional Tone3", ...] } ' ' </pre>

Table 11

Prompt used by the writer agent in the creative tags retrieval.

Creative tags retrieval
<pre> # Goal We will collaboratively engage in theatrical script creation. Prior to this, it is necessary to deconstruct and conceptualize the input script setting. # Setting - ***Core Character Settings***: {characters_str} - ***Fundamental Script Information***: {info_str} - ***Thematic Tone***: {thematic_tone_str} - ***Story Synopsis***: {storyline_str} # Task 1. Based on the 'Setting', design five plot points for each core character. 2. Building upon Task 1, develop three unconventional plot points for each core character. 3. Retrieve as many characters as possible that either match the characteristics of the tags or are associated with them within the given context. # Context "" {label} "" # Output For each character, sequentially output one set of tags per line. Each tag should be an anonymous, subject-omitted summarization phrase, formatted as: (tag_1, tag_2, tag_3, ...) </pre>

Table 12

Prompt used by the writer agent during the new character design stage.

New Character Design
Step one: # Goal As a theatrical script creation system, we collaboratively design characters for the play. Given the director-provided character settings ('Setting'), these constraints ensure alignment with human creative intent. # Character Setting Generate character dossiers for the following roles: {characters_str} # Task 1. Analyze character specifications in 'Setting' to produce dossiers containing: - **base_info**: Gender, age, occupation - **characteristics**: Height/weight, facial features, hairstyle, attire, signature gestureshabits - **personalities**: Primary traits: Dominant qualities Paradoxical nature: Contradictory characteristics (format: "Appears...but actually...") - **background_story**: Family origin, education, lifestyle, interests, beliefs, aspirations - **interpersonal_relationship**: Connections with other key characters (strictly adherent to 'Setting') - **language_style**: Speech patterns, catchphrases, signature lines - **key_events**: Pivotal life experiences, turning points, transformative incidents - **other_detail**: Unique abilities, signature props/objects *Verify content alignment with 'Setting' constraints* 2. Output each character dossier in single-line format: "(role_name)##(base_info)##(characteristics)##(personalities)##(background_story)##(interpersonal_relationship)##(language_style)##(key_events)##(other_detail))##" # Output Step two: # Goal As a theatrical script creation system, we collaboratively design characters. Given creative reference labels ('Labels'), integrate their narrative elements into character dossiers. # Biography Augment and expand dossiers for: {c} # Creative Labels "" {deduplication_labels_str} "" # Character Setting Key character specifications: {characters_str} # Task 1. Analyze current 'Biography' containing: - **base_info**: Gender, age, occupation - **characteristics**: Physique, facial features, hairstyle, attire, signature mannerisms - **personalities**: Dominant traits; Paradoxical nature (format: "Superficially...but intrinsically...") - **background_story**: Lineage, education, lifestyle, interests, beliefs, aspirations - **interpersonal_relationship**: Connections with other key characters - **language_style**: Speech patterns, catchphrases - **key_events**: Critical life milestones, transformative incidents - **other_detail**: Unique abilities, signature props 2. Integrate 'Labels' to supplement: - **background_story**: Life vignettes, interest demonstrations, belief-establishing moments - **interpersonal_relationship**: Potential NPC interactions - **key_events**: Relevant episodic experiences - **other_detail**: Exhaustive imaginative expansions *Verify augmentation aligns with 'Setting' constraints* 3. Output augmented single-line dossier: "(role_name)##(base_info)##(characteristics)##(personalities)##(background_story)##(interpersonal_relationship)##(language_style)##(key_events)##(other_detail))##" # Output Generate only one character dossier per execution.

Table 13
Prompt used by the writer agent during the narrative arc planning stage.

Narrative Arc Planning
Goal Design a narrative arc based on given descriptions and labels, utilizing the Hero's Journey structure as reference. # Real Data *Candidate Creative Labels & Character Biographies*: {context} *Fundamental Script Information*: {info_str} *Thematic Tone*: {thematic_tone_str} *Story Synopsis*: {storyline_str} # Task 1. Analyze provided 'Biographies' to infer character interactions and emotional development 2. Generate initial 'Narrative_arc' based on Task 1: - Each 'Node' represents a single scene with character-centric activities - Split multi-scene nodes into discrete plot points - Ensure node consistency with 'Biographies', 'info_str', 'thematic_tone_str', and 'storyline_str' 3. Augment 'Narrative_arc' using selected labels from 'Labels' 4. Output 'Narrative_arc' in specified format: “(TITLE##CHARACTER_LIST##KEY_EVENT##KEY_ITEM##EMOTIONAL_EXPRESSION##DESCRIPTION)##” # Notification - Maintain character-trait alignment and logical coherence - Leverage occupational attributes for character development - Ensure plot continuity without logical gaps - Preserve emotional expression consistency # Output Narrative arc design: “(Scene Title##Participant Characters##Pivotal Event##Crucial Prop##Emotional Tone##Detailed Description)##”

Table 14

Prompt used by the actor agent during the improvisational dialogue generation stage.

Improvisational Dialogue Generation
Step one: # Goal As a professional theater actor agent, analyze dramatic settings ('Setting') and performance context ('Scene') to decompose scenes into atomic narrative units. # Setting *Core Character Settings*: {characters_str} *Fundamental Script Information*: {info_str} *Thematic Tone*: {thematic_tone_str} *Story Synopsis*: {storyline_str} # Scene {arc_node} *Note: 'Scene' contains current act's title, core participants, key events, crucial props, emotional tone, and detailed description in format: "(TITLE##CHARACTER_LIST##KEY_EVENTS##KEY_ITEMS##EMOTIONAL_EXPRESSION##DESCRIPTION)##" # Biographies Character dossiers: {biographies} # Step 1. Decompose 'Scene' into ≤ 3 atomic events: - Events must be mutually independent and sufficiently simple for multi-turn agent dialogue completion - Employ narrative units: [*linear progression*, *flashback*, *conflict*, *contrast*, *dream/imagery*] to enhance dramatic diversity 2. Per Event, design: - **Characters*: Required participants - **Time*: Temporal setting - **Location*: Spatial setting - Describe either: landmark character action OR group consensus achievement - Validate: Temporal/spatial plausibility Character interaction logic 3. Output format per Event: "(Event: {Description of landmark action/consensus achievement}) ##CHARACTERS##TIME##LOCATION)" # Output Only provide Step 3 results. Step two: # Goal As an actor agent in a multi-turn dialogue generation task, assume character roles and generate performances based on: - High-level script settings ('Setting') - Current performance context ('Scene') - Cast specifications ('Characters') # Setting - ***Story Synopsis***: {storyline_str} - ***Fundamental Script Information***: {info_str} - ***Thematic Tone***: {thematic_tone_str} ## Characters Current speaking participants: - *Narrator*##Provides exposition, explanations, or summaries {characters_str} - *Other NPCs*##Assume random identities to interact with main characters as stochastic narrative elements # Scene Current act description: {arc_node} *Note: 'Scene' format: " (TITLE##CHARACTER_LIST##KEY_EVENT##KEY_ITEM##EMOTIONAL_EXPRESSION##DESCRIPTION)##" containing: Concise title,Core participants,Pivotal events,Crucial props,Emotional tone,Detailed contextual description* # Event Current Dramatic Task Corresponding Event: {event} Remark: Unresolved Issues in Current Performance—{performance_analysis} {bio_desc} # History The multi-turn dialogue generation task has yielded the following historical record: {action_history_str} { # Step 1. Portray "{char_in_turn}"—Execute the performance described in 'Event': - **Core Objective*: Direct the narrative toward event resolution, concluding current performance efficiently - **Current Deficiencies*: {performance_analysis} - **Constraint*: Output precisely one character's performance per response - **Format Requirement*: "Narration (Cinematic Cues)/Character Name (Action/Emotion): Dialogue" - "Narration" may incorporate "cinematic cues" (e.g., "zoom in/out", "close-up", "fade to dark/bright/blur") 2. Evaluate performance completion status against 'Event' criteria: - Output termination flag ('status') and rationale ('reason'): - 'UNFINISHED': Performance incomplete - 'FINISHED': Performance accomplished - 'reason' should follow structured conventions: *Character X achieved Goal A but requires Goal B—suggest next performer address B" or "Event narrative substantially completed—terminate with FINISHED"* 3. Designate next performer ('next_turn'): - For multi-character events: Ensure role alternation (avoid repeating current performer) - Upon 'FINISHED' status: 'next_turn' may be "SCENE_END" # Output Example { "performance": "Narration (Camera slowly pushes in): Outside the cottage, Little Red Riding Hood anxiously awaits Grandma's response", "reason": "Establishing scene shows character's arrival but lacks resolution—suggest next performer as Grandma", "status": "UNFINISHED", "next_turn": "Grandma" }

Table 15
Prompt used by the critic agent during the critique and revision stage.

Critique and Revision
Step one: # Goal You are to assume the role of a professional drama critic. Please review the given draft script and provide a detailed, expert-level analysis and revision suggestions with a focus on plot creativity: - Based on a set of reference creative labels (see 'Labels' below), analyze the "narrative originality" of each scene and suggest potential story beats or visual moments that could be added. Each comment and recommendation should be concise and precise. - Consider whether techniques such as suspense, contrast, flashback, or metaphor could be meaningfully incorporated into the scenes to enhance artistic expression. - Your final output should include "only" revision suggestions related to narrative originality for each scene, formatted as a JSON list: <pre> '''json { "Scene 0: [Revision suggestion]", "Scene 1: [Revision suggestion]", ... } ''' # Labels You may refer to the following creative labels as inspiration: ''' {labels} ''' </pre> Step two: # Goal You are to act as a professional drama critic. Based on the provided revision suggestions, please revise the given script excerpt accordingly. # Narrative Arc The overarching narrative arc of the script is as follows: <pre> {arc_text} </pre> # Scene The script excerpt to be revised is as follows: <pre> ''' {s} ''' </pre> # Suggestion Please revise the script according to the following suggestions: <pre> ''' {sugg} ''' </pre> # Attention Ensure that your output strictly follows the format below. Do not include any additional explanation or commentary: <pre> ''' # Scene N: Retain the original context and situational setup as described in the given 'Scene'. ## Storyboard: Preserve the original storyboard as described in the given 'Scene'. For any newly added shots, use the format: "## Shot Description ##Character List##Time##Location" Dialogue: [Your revised sequence of character dialogues] ... ''' </pre> # Output (Only output the revised scene dialogues. Do not include any extra explanation or commentary.)

Table 16
Prompt for generating creative tags and suggestions.

Creative tags and suggestions generation
Goal You are a creative screenwriter, skilled at providing innovative ideas and suggestions. Given a summary of the previous scenes, the setting, the narrative arc, and the scene dialogues of a screenplay, you need to derive relevant creative tags based on the current scene and generate creative suggestions for each scene. ## Previous Scenes <pre> {prescene} </pre> ## Setting <pre> {setting} </pre> ## Narrative Arc <pre> {narrative_arc} </pre> ## Scene Dialogue <pre> {scene_dialogue} </pre> ## Step 1. Identify 'Focus': Identify the thematic focus and character performance focus of each scene. 2. Construct 'Query': Build a query based on the summary of the previous scenes and the 'focus' of the current scene. 3. Retrieve 'Creative Tags': Use the 'query' to retrieve potentially useful creative tags. 4. Generate 'Creative Suggestions': Create structured creative suggestions for the current scene based on the retrieved 'creative tags'. 5. Output Results: Output the formatted 'creative tags' and 'creative suggestions'. ## Output Format <pre> (creative tag1, creative tag2, creative tag3, ...) (Creative Suggestions) </pre>

Appendix C. Corpus of classical scripts

We provide the complete list of the 48 classical drama scripts used in our study. As described in [Section 2](#), these scripts were collected from publicly available Chinese-language platforms, reviewed by invited directors, and curated to ensure both thematic alignment and narrative diversity. The corpus covers a wide range of genres, including historical drama, sketch comedy, and straight play, and serves as the foundation for the ontology-based processing methods detailed in [Section 3](#). Presenting the corpus in full enhances the transparency and reproducibility of our experimental setup, while also illustrating the narratological and structural variety on which our evaluations are grounded. [Table 17](#) presents the corpus of Classical Scripts employed in this co-creation; as the original material is in Chinese, we provide translated titles within the table.

Table 17
Corpus of classical scripts.

ID	Title	Genres labels	Words	Synopsis
1	The Pothole	Modern Drama, Sketch Comedy, Straight Play	1101	Dongjiao Suburb, Xihong City: a gaping road crater keeps injuring commuters. Street-office director Zhu passes the buck, but newly arrived chief Ma's undercover visit exposes the neglect. Zhu is removed for gross dereliction, underscoring a new zero-tolerance stance on official inertia.
2	Lighting the Lamp	Modern Drama, Sketch Comedy, Straight Play	1279	As Village Chief Liu's term ended, the villagers deliberately delayed turning on the streetlights to persuade her to stay, grateful for her leadership in poverty alleviation. Chief Liu ultimately decided to remain for another three years, with her husband expressing support, reflecting the deep bond between the community and its leader.
3	Youth Aspiration	Modern Drama, Time-Travel Drama, Full-Length Multi-Act Play, Straight Play	1887	At the exhibition hall, guide Gong encountered the time-traveling hero Du . Through their conversation, Du's spirit of sacrifice was revealed. Learning that his comrades were safe, Du returned to his original time, while a collective declaration highlighted the youth's dedication to striving for a better future.
4	Uncle Zhang	Modern Drama, Full-Length Multi-Act Play, Straight Play	1889	University student Yan returned to her hometown for research. Her sister was working on a wellness town project but faced obstacles when Elder Zhang refused to hand over his land due to sentimental attachment. Through patient communication, Elder Zhang gradually understood the significance of rural revitalization, resolving the conflict.
5	Dream Chaser	Modern Drama, Sketch Comedy, Musical Theatre	1883	Three generations of workers from one family gathered at the factory gate for a celebration. The grandfather reminisced about industrial changes, while his children and grandchildren looked forward to a technological future. Their joint singing expressed the shared ideals of workers in the new era.
6	I Remember You, You Will Live On	Historical Drama, Time-Travel Drama, Full-Length Multi-Act Play, Straight Play	2159	In 1949, Wang time-traveled to the Shanghai Liberation Memorial Hall in 2021, where she learned about her own sacrifice and chose to return to the battlefield. The guide paid a tearful tribute, conveying the enduring theme of revolutionary spirit.
7	From Past to Present in Daigu	Modern Drama, Full-Length Multi-Act Play, Straight Play	2089	In 1980, university student Da tried to bring his girlfriend to his hometown but couldn't reach it due to rugged mountain roads. By 2024, villagers were discussing long-term development topics like their children's education. An entrepreneur conducting a survey proposed integrating culture and tourism, bringing new hope for the hometown's future.
8	New Melody of Aishan	Modern Drama, Musical Theatre, Full-Length Multi-Act Play	2329	Ai, the first secretary of the village, returned to her hometown to promote the cultivation of mugwort. At first, Er, a villager, was unwilling to hand over his land. After multiple patient communications with Ai, Er gradually understood the development prospects of mugwort, took the initiative to join the cultivation, and jointly promoted the industrial transformation of his hometown.
9	My Heart Soars	Modern Drama, Full-Length Multi-Act Play, Straight Play	2512	Sun, a female pilot, was selected as a reserve team member due to her all-round excellence. She was initially unwilling to accept this. After being enlightened by the team leader, she understood the mission of a reserve team member. Her teammates wrote her name on their flight suits, jointly forging the team's honor.
10	The Choice	Modern Drama, Full-Length Multi-Act Play, Straight Play	2515	Village Chief Wei tried live-streaming to sell tea to solve the problem of unsold tea. He encountered setbacks in the early stage and was questioned by his girlfriend. However, he persisted and gradually achieved results, demonstrating the hardships and hopes of rural revitalization.
11	Military Spirit	Modern Drama, Sketch Comedy, Straight Play	2396	Jun had to return to his post for an emergency mission during a home visit, and unfortunately made the ultimate sacrifice heroically in the fight against floods and rescue work. When the leader came to express condolences, his mother and wife held back their grief and encouraged his child to grow up to be a brave and responsible person like his father, passing on this responsibility and love.
12	Who Doesn't Praise Our Beautiful Hometown	Modern Drama, Sketch Comedy, Musical Theatre	2247	There were representatives from 20 towns in total. They showcased the characteristics of each town through cross-talk, poetry recitation, and clapper talk respectively. Finally, they gathered together to pay New Year greetings, praise the beauty of their hometowns, and send New Year blessings.
13	The Special Party Fee	Modern Drama, Full-Length Multi-Act Play, Straight Play	2653	During a group activity, Li was reluctant to contribute, while Wang, despite his own financial struggles, generously donated. After being interviewed by a journalist, it was revealed that Wang had personally experienced disaster relief efforts and understood the hardships of receiving aid. His actions and story deeply moved Li, who felt ashamed and finally grasped the true weight of selfless dedication.
14	On the Road to Rural Revitalization	Modern Drama, Sketch Comedy, Straight Play	2620	Manager Xie's crab farm suffered typhoon damage, leaving him unable to pay rent. While Director Zhou pressed for payment, Village Chief Zhang applied for subsidies and even covered the rent temporarily. Initially misunderstood by the villagers, Chief Zhang later earned their gratitude as everyone joined together in hoping for better days ahead.

Table 17
Continued.

ID	Title	Genres labels	Words	Synopsis
15	Spring Rain	Modern Drama, Sketch Comedy, Straight Play	2740	An argument broke out when an elderly couple tried to obtain a replacement marriage certificate for their son. Qing, the village cadre, visited their home and informed them that the village service station could handle the procedure while also providing agricultural supplies. Like a welcome rain after a long drought, the solution arrived just when it was needed most.
16	The Ribbon-Cutting	Modern Drama, Sketch Comedy, Straight Play	2754	A couple prepared to move into their new house and planned an elaborate ribbon-cutting ceremony for their pig farm, intending to invite township leaders with gifts and banquets. However, Township Chief Zhang called to decline the ceremony, urging them to invest the money into production instead. The couple agreed with this practical suggestion.
17	The Torch of Chang'an	Modern Drama, Full-Length Multi-Act Play, Straight Play	2942	Tao's mother helped an elderly man transfer money. When Tao tried to use his homework as an excuse to receive snacks, his grandfather shared a story about refusing gifts and approving loans to teach him a lesson. On Tao's birthday, his mother returned to work, passing on the family's values of integrity and responsibility.
18	Liu Hulan	Historical Drama, Full-Length Multi-Act Play	2846	A mother and her child visited Liu's tomb to pay their respects. The mother told the child that in 1947, Liu was arrested. She refused the temptation from Zhang and sacrificed her life heroically. This memory of courage and dignity has been deeply engraved in people's hearts and passed down from generation to generation.
19	From the Pages of History	Historical Drama, Straight Play, Full-Length Multi-Act Play	3030	A six-act play condenses a century of vicissitudes. It starts from the turbulent days of the old times, follows the awakening and struggle of people with lofty ideals, goes through hard times, and finally ushers in a new life. Generations of young people have stepped forward and continued to strive, demonstrating a great journey of self-improvement.
20	Poverty Alleviation Storm	Modern Drama, Full-Length Multi-Act Play, Straight Play	3102	Village Chief Yang visited Dong's home. He found Dong addicted to alcohol and eating the chickens provided for poverty alleviation. Village Chief Yang advised Dong to grow flue-cured tobacco, but Dong expressed doubts. His son's dissatisfaction prompted a change in Dong, and finally, Dong's family made up their minds to shake off poverty.
21	Xiao Jiu	Modern Drama, Sketch Comedy	3092	Le was henpecked by his wife Jiu, yet he showed his love for her in every aspect of their family life. Jiu suspected that Le had borrowed money behind her back, which led to constant quarrels between them. In the end, Le confessed that he had lent money to his ex-girlfriend to help her out of an emergency. Jiu understood and supported him, and their marital relationship became even stronger.
22	Farewell Across a Thousand Miles	Modern Drama, Straight Play	3061	During the COVID-19 pandemic, Dr. Li and Gui, who were doctors at a hospital in Wuhan, entrusted their child to their parents in their hometown and resolutely returned to work to safeguard lives. Although the parents were reluctant to part with them, they finally understood and supported their decision, demonstrating the sense of responsibility, love, and dedication of medical workers.
23	Fancy But Empty	Modern Drama, Sketch Comedy	3287	During his inspection visits, Mayor Lei discovered that some business owners had set up unnecessary decorations to cope with inspections, which affected their normal operations. After understanding the situation, he promoted the simplification of requirements and optimization of management. The business owners also removed the formalistic decorations one after another and shifted their focus back to daily operations.
24	Be the "Nail" in Rural Revitalization	Modern Drama, Straight Play	3489	Wei, a village cadre, led the villagers of Dabai Village to open up new sales channels for tea through e-commerce live streaming. After experiencing initial setbacks, they finally achieved success with the support of various parties, effectively increasing the villagers' income and promoting the development of their hometown.
25	The Devils Enter Wanlian	Historical Drama, Sketch Comedy	3398	In 1937 North China, Li risked his reputation by disguising his identity to protect his fellow villagers, skillfully navigating interactions with the occupiers. In the end, he revealed his true self and rose up in resistance.
26	Who Will Be the Live Streamer	Modern Drama, Sketch Comedy	3664	The village chief and Alan were searching for a live-stream host for an agricultural product exhibition but faced difficulties when a celebrity host canceled last minute. Ultimately, Kunzai, who had been underestimated by the villagers, unexpectedly stepped up as the host and successfully promoted sales using his new media skills.
27	The Hourly Worker	Modern Drama, Sketch Comedy	3622	Zhao felt lonely struggling to adapt to city life, so his son hired Song as a companion. Through a series of misunderstandings and conversations, Zhao eventually realized that elderly people need to find their own joy and face life positively.
28	The General and the Soldier	Modern Drama, Straight Play	2971	Faced with training setbacks, Hou, spurred on by Corporal Li and inspired by his father's advice to "stay true to himself," gained a deep understanding of what it means to be a leader. Through concrete actions, he achieved a personal breakthrough, demonstrating dedication and integrity.

Table 17
Continued.

ID	Title	Genres labels	Words	Synopsis
29	The Village Official's Wife	Modern Drama, Sketch Comedy	3718	After returning to her rural hometown, university graduate Erxia actively pushed for change and founded a home for left-behind children. Despite initial misunderstandings from her family and fellow villagers, she moved them with her sincerity and results, eventually gaining their full support and showcasing the perseverance and ideals of the younger generation.
30	I Want a Home	Modern Drama, Sketch Comedy	3892	Zhao and Li recorded a matchmaking advertisement at a TV station, helping each other practice and sharing their life experiences. In the end, they expressed their longing for family through a duet of "I Want a Home," reflecting the challenges faced by middle-aged singles in finding love and their pursuit of genuine companionship.
31	The Heart's Draw	Modern Drama, Straight Play	3872	During the farming off-season, Uncle Liu and Uncle Wang argued over the allocation of two new houses. To ensure fairness, Village Chief Ma quietly gave up his own allocation. When the two men learned this from the housing list, they were deeply moved. The village chief's selflessness resolved the dispute and earned everyone's respect.
32	The Xi'an Incident	Historical Drama, Full-Length Multi-Act Play, Straight Play	3953	In 1936, after multiple unsuccessful attempts to persuade Chiang Kai-shek to unite against the enemy, Zhang and Yang took action. With the efforts and mediation of various parties, the incident was finally resolved peacefully, creating an opportunity for comprehensive unity in the fight against the enemy.
33	The Planner	Modern Drama, Sketch Comedy, Straight Play	3783	Bai and Hei's family rooster laid eggs, and Niu intended to hype this up. After Bai signed a contract, the rooster was stewed. Hei cleverly dealt with the situation using the stewed chicken and satirized the chaos of hype with absurd advertising slogans.
34	A Sycamore Tree in the Courtyard Attracts Golden Phoenixes	Modern Drama, Sketch Comedy, Straight Play	3015	Aunt Wang argued with her daughter-in-law Fang because her poultry went missing. Fang explained that she had handled the poultry in response to the "Beautiful Courtyard" initiative. Later, Fang's family was awarded the title of "Model Household," which inspired Aunt Li to also decide to improve her own courtyard.
35	To Help or Not to Help	Modern Drama, Sketch Comedy, Straight Play	4003	Hao helped an elderly aunt who fell down, but was mistakenly accused of knocking her over. With the testimony of passers-by and clarification from the police, the aunt learned the truth. Hao conveyed the positive message that "people's kindness and integrity must never falter."
36	Yuan Xiang	Modern Drama, Full-Length Multi-Act Play, Musical Theatre	3929	Yuan, a college student, wanted to give back to her hometown with her knowledge. Although her idea of setting up an art class was opposed by her mother, it was affirmed by the villagers. With the help of her classmate Liu, their cultural and tourism cooperative successfully got off the ground, illuminating the familiar streets and lanes with art.
37	Class Reunion	Modern Drama, Sketch Comedy, Straight Play	4387	Fan prepared for a classmates' reunion, hoping to curry favor with President Wu. He mistakenly took his old classmate Zhao for a porter. Zhao revealed that he was both President Wu and the husband of Wang, and pointed out that true feelings should be cherished at classmates' reunions.
38	Heart Disease	Modern Drama, Sketch Comedy, Straight Play	4388	Fan was so excited about winning three million yuan in the lottery that he fell ill. His wife took him to Zhao for treatment. Zhao used a "collapse therapy" approach, during which Fan confessed that his extramarital partner was Zhao's wife. Later, when Fan was distributing the money, Zhao was so excited that he fainted.
39	Man Jiang Red	Historical Drama, Full-Length Multi-Act Play, Straight Play, Time-Travel Drama	4777	A time-travel experience at the museum allowed the children to meet a young hero from history. Listening to him talk about how people in those days used their respective strengths to defend their homeland taught the modern children a profound lesson about courage and responsibility.
40	Yesterday, Today and Tomorrow	Modern Drama, Sketch Comedy, Straight Play	4596	Hei and Bai were guests on the program Tell It Like It Is. They talked about their past experience of "collecting small gains" and their current marital crisis of living separately, then shared their visions for the future. They humorously demonstrated the changes in rural areas and the affection between husband and wife.
41	The Water Delivery Man	Modern Drama, Sketch Comedy, Straight Play	4823	Zhao temporarily played the role of a "stepfather" to help Gao cover up a lie when her son Fan, who had returned from studying abroad, came to visit. During their conversation over drinks, the truth gradually emerged. Zhao finally couldn't bear to keep lying and told Fan about the hardships his mother had endured to support his education alone. Mother and son reunited, and the story came to an end with warmth and humor.
42	Paying a New Year Call	Modern Drama, Sketch Comedy, Straight Play	5044	Zhao and his wife took gifts to visit Village Head Fan in the hope of contracting a fishpond. Unexpectedly, Fan had already stepped down from his position, and the couple's attitude immediately turned cold. However, Fan told them that they could still contract the fishpond. This little mix-up fully revealed the warmth and coldness of human relationships and the subtle changes in worldly affairs.
43	Talking About Things	Modern Drama, Sketch Comedy, Straight Play	5078	Bai and Hei appeared on the talk show Xiaocui's Chats, where Bai boasted about her rise to fame while Hei repeatedly exposed her exaggerations—revealing that her "private jet" was actually a tractor and her "mink coat" was rented. The awkwardness was eventually diffused by their authentic Errenzhuan performance, showcasing their genuine, down-to-earth nature.

Table 17
Continued.

ID	Title	Genres labels	Words	Synopsis
44	Donation	Modern Drama, Sketch Comedy, Straight Play	6350	Qian accidentally donated 15,000 yuan belonging to his in-law Bai, becoming an “online hero.” During a live stream, Bai exposed the truth, but the widow who received the donation offered to return the money. The story unfolds as a farce dubbed “donating a father,” exploring themes of integrity and social image.
45	Not Short of Money	Modern Drama, Sketch Comedy, Straight Play	6578	Zhao took his granddaughter Yadan to dinner with Bi in hopes of securing a spot on Star Avenue. The waiter, Shen, unexpectedly showcased his talent, and both ended up getting an opportunity. The story concluded humorously with the line, “My grandpa’s surname is also Bi.”
46	Green Mountains and Clear Waters	Modern Drama, Full-Length Multi-Act Play, Straight Play	5637	Overwhelmed by city life, Bai returned to her hometown only to find its ecosystem severely damaged. She led the villagers in environmental restoration, recovering the green mountains and clear waters, reconciled with her father, and rediscovered her purpose in her rural roots.
47	Wild Boar Nest Village	Modern Drama, Full-Length Multi-Act Play, Straight Play	7707	Yezhuwo Village once struggled with wild boar infestations. After Guo took over as the new director, he invited agricultural experts to guide the villagers in switching to high-value medicinal herbs. Gradually, the village overcame poverty. As life improved, the village was renamed “Zhenxing Village,” symbolizing a new lease on life for the community.
48	The Declaration Will Never Die	Historical Drama, Full-Length Multi-Act Play, Straight Play	27,317	While pursuing a cause for justice, Lin ended up in danger and sought refuge in the home of her ex-husband Kuang, causing turmoil in the family. Their son, Kuang, was deeply moved after learning the truth. Despite his efforts to rescue her, he ultimately failed. In the face of persecution, Kuang set his house on fire as a final act of defiance. Although Lin did not survive, the light and ideals she pursued live on in the next generation.

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